Story 3

Pricilla

2026-03-05

r setup, include=FALSE knitr::opts_chunk$set(echo = TRUE)

```r
install.packages("dplyr")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```r
install.packages("readxl")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```r
install.packages("ggplot2")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```r
install.packages("tidyverse")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```r
install.packages("tidyr")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(readxl)
library(ggplot2)
library(forcats) # For fct_reorder
library(patchwork)
library(tidyr)
```

```r
# URLs
gun_laws_url <- "https://raw.githubusercontent.com/prnakyazze94/Data-608/refs/heads/main/Gun_laws"
mortality_url <- "https://raw.githubusercontent.com/prnakyazze94/Data-608/refs/heads/main/Mortality__by
```

```
gun_mortalityData_url <- "https://raw.githubusercontent.com/prnakyazze94/Data-608/refs/heads/main/Gun_M

# Read data
gun_laws <- read.csv(gun_laws_url, stringsAsFactors = FALSE)
mortality_by_years <- read.csv(mortality_url, stringsAsFactors = FALSE)
gun_mortalityData <- read.csv(gun_mortalityData_url, stringsAsFactors = FALSE)

# View structure
str(gun_laws)
```

```
## 'data.frame':    50 obs. of  4 variables:
##  $ Series                         : chr  "State" "State" "State" "State" ...
##  $ State                          : chr  "AL" "AK" "AZ" "AR" ...
##  $ Strength.of.Gun.Laws           : num  12 9 7.5 4.5 90.5 67.5 81.5 65 28 5 ...
##  $ Gun.Deaths.per.100.000.Residents: num  25.6 23.5 18.5 21.9 8 16.6 6.2 12 13.7 18.6 ...
```

```
str(mortality_by_years)
```

```
## 'data.frame':    358 obs. of  9 variables:
##  $ X                 : chr  "" "" "" "" ...
##  $ State             : chr  "Alabama" "Alabama" "Alabama" "Alabama" ...
##  $ State.Code        : int  1 1 1 1 1 1 1 2 2 2 ...
##  $ Year              : num  2018 2019 2020 2021 2022 ...
##  $ Year.Code         : int  2018 2019 2020 2021 2022 2023 NA 2018 2019 2020 ...
##  $ Deaths            : int  1064 1076 1141 1315 1278 1292 7166 155 179 175 ...
##  $ Population        : int  4887871 4903185 4921532 5039877 5074296 5108468 29935229 737438 731545 7
##  $ Crude.Rate        : num  21.8 21.9 23.2 26.1 25.2 25.3 23.9 21 24.5 23.9 ...
##  $ X..of.Total.Deaths: chr  "0.4%" "0.4%" "0.4%" "0.5%" ...
```

```
state_lookup <- data.frame(
  Label = state.abb,       # abbrev
  State = state.name,      # full name
  stringsAsFactors = FALSE
)
```

```
# Clean column names (remove spaces and special characters)
colnames(gun_laws) <- make.names(colnames(gun_laws))

# Create Likert scale (1-5)
gun_laws <- gun_laws %>%
  mutate(
    Strength.of.Gun.Laws = as.numeric(Strength.of.Gun.Laws),
    gun_law_strength = cut(
      Strength.of.Gun.Laws,
      breaks = c(0, 20, 40, 60, 80, 100),
      labels = c("1", "2", "3", "4", "5"),
      include.lowest = TRUE,
      right = FALSE
    )
  )

# View result
print(gun_laws)
```

```
##    Series State Strength.of.Gun.Laws Gun.Deaths.per.100.000.Residents
```

```
## 1   State   AL                 12.0                              25.6
## 2   State   AK                  9.0                              23.5
## 3   State   AZ                  7.5                              18.5
## 4   State   AR                  4.5                              21.9
## 5   State   CA                 90.5                               8.0
## 6   State   CO                 67.5                              16.6
## 7   State   CT                 81.5                               6.2
## 8   State   DE                 65.0                              12.0
## 9   State   FL                 28.0                              13.7
## 10  State   GA                  5.0                              18.6
## 11  State   HI                 80.0                               4.9
## 12  State   ID                  3.5                              17.9
## 13  State   IL                 85.5                              13.5
## 14  State   IN                 18.0                              18.3
## 15  State   IA                 15.0                              10.5
## 16  State   KS                  9.5                              16.3
## 17  State   KY                  9.0                              18.4
## 18  State   LA                 12.5                              28.3
## 19  State   ME                 27.0                              14.0
## 20  State   MD                 78.5                              12.3
## 21  State   MA                 86.5                               3.7
## 22  State   MI                 37.0                              13.9
## 23  State   MN                 53.5                               8.9
## 24  State   MS                  4.0                              29.4
## 25  State   MO                  8.0                              21.4
## 26  State   MT                  5.0                              21.5
## 27  State   NE                 23.5                              10.6
## 28  State   NV                 37.0                              18.4
## 29  State   NH                  9.5                               9.6
## 30  State   NJ                 80.5                               4.6
## 31  State   NM                 42.0                              25.3
## 32  State   NY                 85.0                               4.7
## 33  State   NC                 24.5                              16.4
## 34  State   ND                 11.5                              12.8
## 35  State   OH                 14.0                              15.0
## 36  State   OK                  7.5                              19.9
## 37  State   OR                 65.0                              14.2
## 38  State   PA                 39.5                              13.6
## 39  State   RI                 60.5                               4.8
## 40  State   SC                 14.0                              19.1
## 41  State   SD                  5.5                              12.3
## 42  State   TN                 14.0                              22.0
## 43  State   TX                 13.5                              14.9
## 44  State   UT                 11.0                              14.8
## 45  State   VT                 41.0                              12.0
## 46  State   VA                 49.0                              13.8
## 47  State   WA                 71.5                              13.0
## 48  State   WV                 18.5                              16.8
## 49  State   WI                 27.0                              12.7
## 50  State   WY                  6.5                              21.5
##    gun_law_strength
## 1                 1
## 2                 1
## 3                 1
```

```
## 4                    1
## 5                    5
## 6                    4
## 7                    5
## 8                    4
## 9                    2
## 10                   1
## 11                   5
## 12                   1
## 13                   5
## 14                   1
## 15                   1
## 16                   1
## 17                   1
## 18                   1
## 19                   2
## 20                   4
## 21                   5
## 22                   2
## 23                   3
## 24                   1
## 25                   1
## 26                   1
## 27                   2
## 28                   2
## 29                   1
## 30                   5
## 31                   3
## 32                   5
## 33                   2
## 34                   1
## 35                   1
## 36                   1
## 37                   4
## 38                   2
## 39                   4
## 40                   1
## 41                   1
## 42                   1
## 43                   1
## 44                   1
## 45                   3
## 46                   3
## 47                   4
## 48                   1
## 49                   2
## 50                   1
```

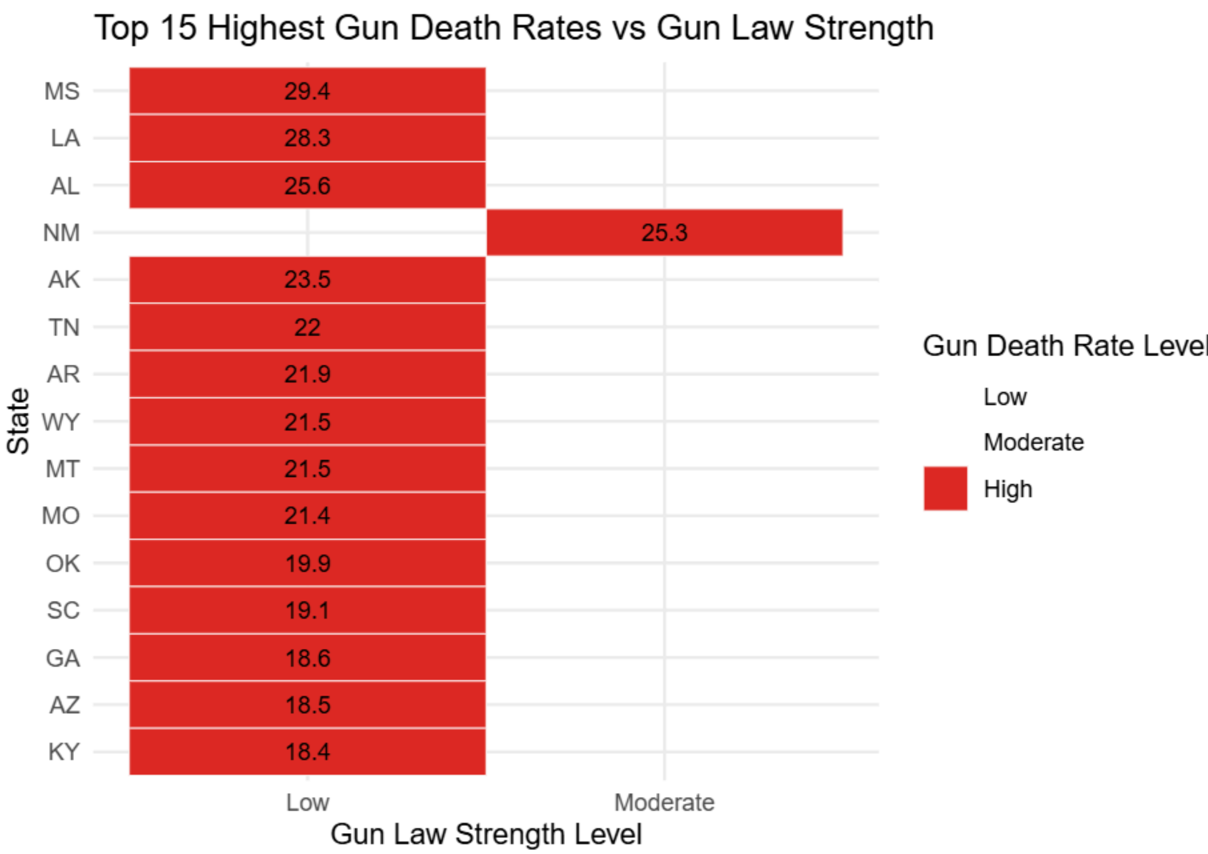
```r
gun_laws_full <- gun_laws %>%
  left_join(state_lookup, by = "State") %>%
  mutate(
    gun_law_strength = as.numeric(gun_law_strength),
    Gun.Deaths.per.100.000.Residents = as.numeric(Gun.Deaths.per.100.000.Residents)
  )
```

```
gun_laws_full <- gun_laws_full %>%
  mutate(
    death_level = cut(
      Gun.Deaths.per.100.000.Residents,
      breaks = unique(quantile(Gun.Deaths.per.100.000.Residents,
                               probs = c(0, .33, .66, 1), na.rm = TRUE)),
      labels = c("Low", "Moderate", "High")[1:(length(unique(quantile(Gun.Deaths.per.100.000.Residents,
                                                                     probs = c(0, .33, .66, 1), na.r
      include.lowest = TRUE
    ),

    law_level = cut(
      gun_law_strength,
      breaks = unique(quantile(gun_law_strength,
                               probs = c(0, .33, .66, 1), na.rm = TRUE)),
      labels = c("Low", "Moderate", "High")[1:(length(unique(quantile(gun_law_strength,
                                                                    probs = c(0, .33, .66, 1), na.rm
      include.lowest = TRUE
    )
  )
```

```
gun_laws_full <- gun_laws_full %>%
  mutate(
    death_level = cut(
      Gun.Deaths.per.100.000.Residents,
      breaks = unique(quantile(Gun.Deaths.per.100.000.Residents,
                               probs = c(0, .33, .66, 1), na.rm = TRUE)),
      labels = c("Low", "Moderate", "High")[1:(length(unique(quantile(Gun.Deaths.per.100.000.Residents,
                                                                    probs = c(0, .33, .66, 1), na.rm
      include.lowest = TRUE
    ),

    law_level = cut(
      gun_law_strength,
      breaks = unique(quantile(gun_law_strength,
                               probs = c(0, .33, .66, 1), na.rm = TRUE)),
      labels = c("Low", "Moderate", "High")[1:(length(unique(quantile(gun_law_strength,
                                                                    probs = c(0, .33, .66, 1), na.rm
      include.lowest = TRUE
    )
  )
```

```
# Filter for Top 15
top_15_data <- gun_laws_full %>%
  arrange(desc(Gun.Deaths.per.100.000.Residents)) %>%
  slice_head(n = 15)

# Plot Top 15
ggplot(top_15_data,
       aes(x = law_level,
           y = fct_reorder(State, Gun.Deaths.per.100.000.Residents))) +

  geom_tile(aes(fill = death_level), color = "white") +
  geom_text(aes(label = round(Gun.Deaths.per.100.000.Residents, 1)), size = 3) +
```

```r
  scale_fill_manual(
    values = c("Low" = "#fee5d9", "Moderate" = "#fcae91", "High" = "#de2d26"),
    drop = FALSE # Keeps all legend items even if not present in these 15
  ) +

  labs(
    title = "Top 15 Highest Gun Death Rates vs Gun Law Strength",
    x = "Gun Law Strength Level",
    y = "State",
    fill = "Gun Death Rate Level"
  ) +

  theme_minimal()
```

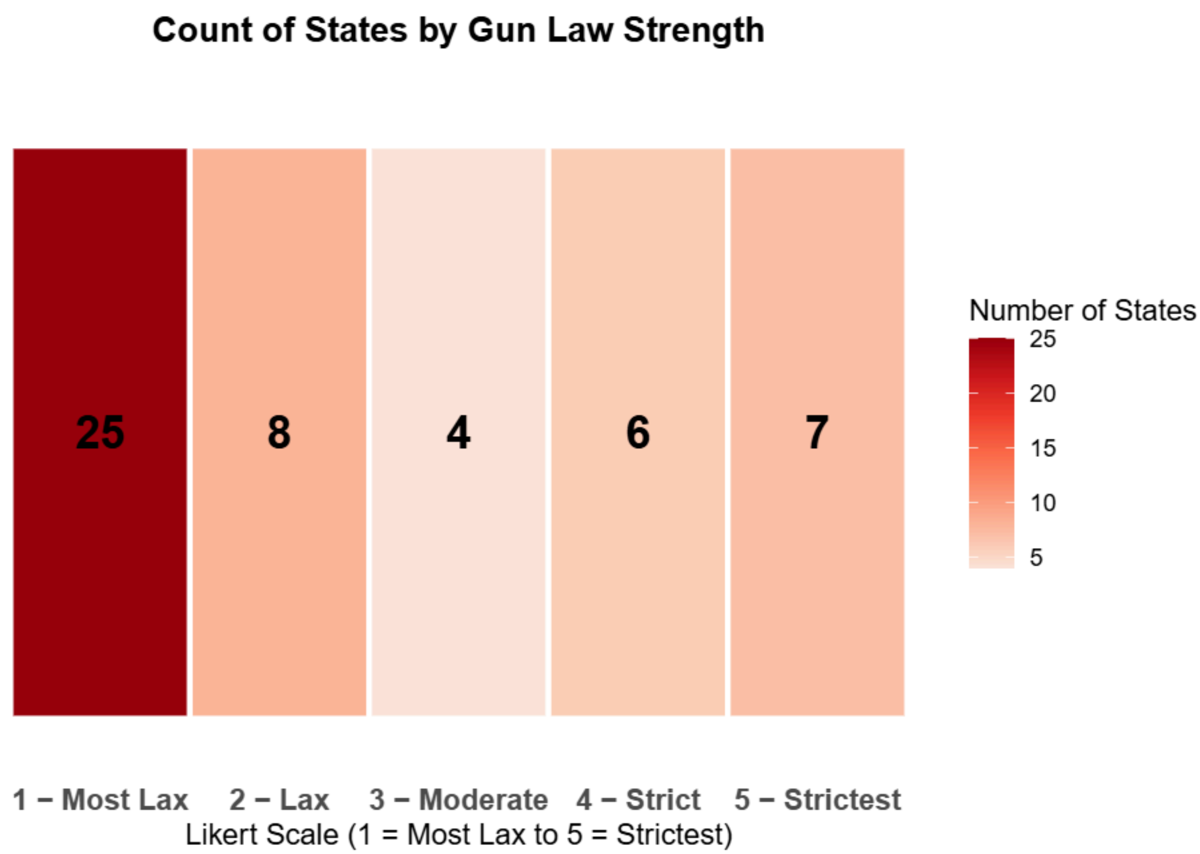



```r
gun_laws_full <- gun_laws_full %>%
  mutate(
    # cut_interval divides the scoring range into 5 equal sections
    law_likert_5pt = cut_interval(
      gun_law_strength,
      n = 5,
      labels = c(
        "1 - Most Lax",
        "2 - Lax",
        "3 - Moderate",
        "4 - Strict",
        "5 - Strictest"
```

```r
      )
    )
  )

# Check how many states landed in each bin
table(gun_laws_full$law_likert_5pt)
```

```
##
##  1 - Most Lax        2 - Lax  3 - Moderate    4 - Strict 5 - Strictest
##            25              8             4             6             7
```

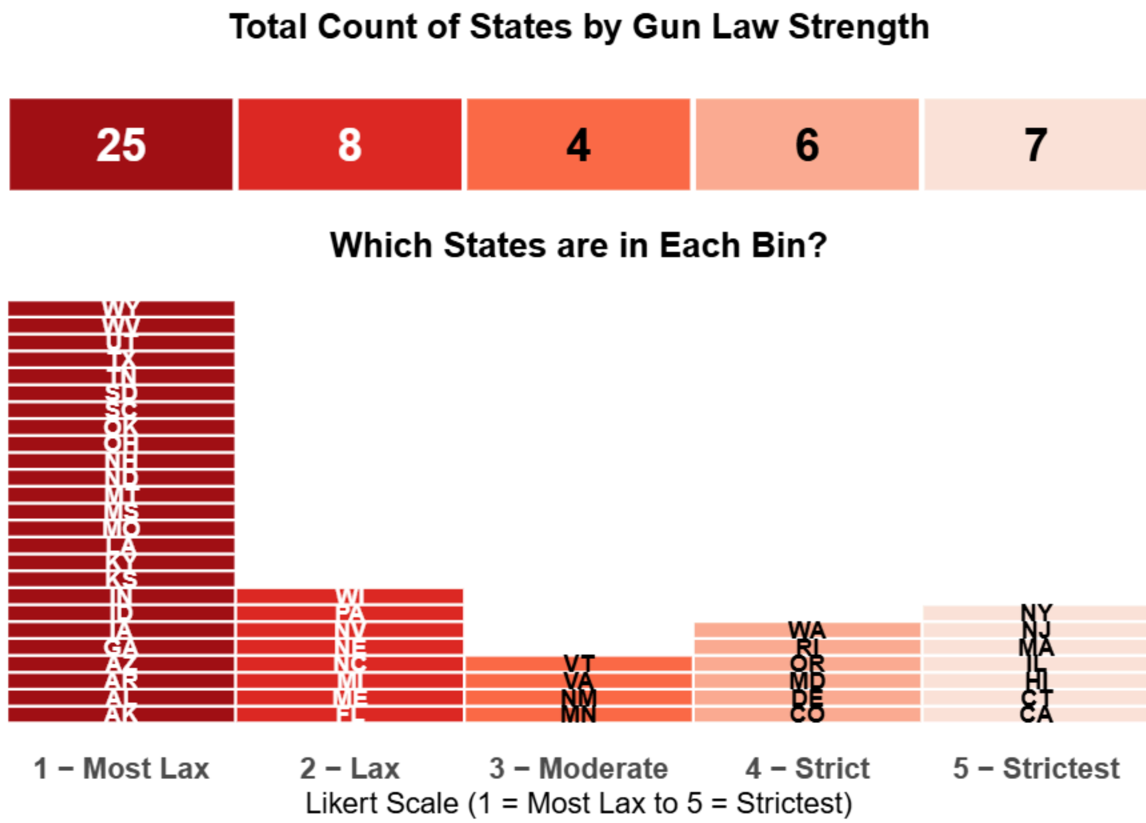
```r
# 1. Manually create a dataframe from your table results
likert_summary <- data.frame(
  Likert_Bin = c("1 - Most Lax", "2 - Lax", "3 - Moderate", "4 - Strict", "5 - Strictest"),
  State_Count = c(25, 8, 4, 6, 7)
)

# 2. Lock the factor levels so they plot in the correct 1-to-5 order
likert_summary$Likert_Bin <- factor(likert_summary$Likert_Bin, levels = likert_summary$Likert_Bin)

# 3. Build the Heatmap
ggplot(likert_summary, aes(x = Likert_Bin, y = "All States", fill = State_Count)) +

  # geom_tile draws the heatmap squares
  geom_tile(color = "white", linewidth = 1) +

  # Add the exact count of states inside each square
  geom_text(aes(label = State_Count), color = "black", size = 6, fontface = "bold") +

  # Apply a color gradient (darker red = more states)
  scale_fill_distiller(palette = "Reds", direction = 1) +

  labs(
    title = "Count of States by Gun Law Strength",
    x = "Likert Scale (1 = Most Lax to 5 = Strictest)",
    y = NULL,
    fill = "Number of States"
  ) +

  theme_minimal() +
  theme(
    panel.grid = element_blank(),      # Remove background gridlines
    axis.text.y = element_blank(),     # Hide the dummy "All States" text on the Y axis
    plot.title = element_text(hjust = 0.5, face = "bold", margin = margin(b = 15)),
    axis.text.x = element_text(size = 11, face = "bold")
  )
```




```r
# ==========================================
# 1. BUILD THE SUMMARY PLOT
# ==========================================
likert_summary <- data.frame(
  Likert_Bin = c("1 - Most Lax", "2 - Lax", "3 - Moderate", "4 - Strict", "5 - Strictest"),
  State_Count = c(25, 8, 4, 6, 7)
)
likert_summary$Likert_Bin <- factor(likert_summary$Likert_Bin, levels = likert_summary$Likert_Bin)

summary_plot <- ggplot(likert_summary, aes(x = Likert_Bin, y = "All States", fill = Likert_Bin)) +
  geom_tile(color = "white", linewidth = 1) +

  # Text turns white on dark bins (1, 2) and black on light bins
  geom_text(
    aes(
      label = State_Count,
      color = ifelse(Likert_Bin %in% c("1 - Most Lax", "2 - Lax"), "white", "black")
    ),
    size = 6,
    fontface = "bold"
  ) +

  scale_fill_brewer(palette = "Reds", direction = -1) +
  scale_color_identity() +

  labs(
    title = "Total Count of States by Gun Law Strength",
```

```
    x = NULL,
    y = NULL
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    panel.grid = element_blank(),
    axis.text.y = element_blank(),
    plot.title = element_text(hjust = 0.5, face = "bold", margin = margin(b = 15)),
    axis.text.x = element_blank()
  )

# ==========================================
# 2. BUILD THE DETAILED STATE PLOT
# ==========================================
detailed_bins <- gun_laws_full %>%
  filter(!is.na(law_likert_5pt)) %>%
  group_by(law_likert_5pt) %>%
  arrange(State) %>%
  mutate(stack_pos = row_number()) %>%
  ungroup()

state_plot <- ggplot(detailed_bins, aes(x = law_likert_5pt, y = stack_pos)) +
  geom_tile(aes(fill = law_likert_5pt), color = "white", linewidth = 0.5) +

  # UPDATE: Removed toupper(), just using the standard 'State' column
  geom_text(
    aes(
      label = State,
      color = ifelse(law_likert_5pt %in% c("1 - Most Lax", "2 - Lax"), "white", "black")
    ),
    size = 3,
    fontface = "bold"
  ) +

  scale_fill_brewer(palette = "Reds", direction = -1) +
  scale_color_identity() +

  labs(
    title = "Which States are in Each Bin?",
    x = "Likert Scale (1 = Most Lax to 5 = Strictest)",
    y = NULL
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    panel.grid = element_blank(),
    axis.text.y = element_blank(),
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.text.x = element_text(size = 11, face = "bold")
  )

# ==========================================
```

```r
# 3. COMBINE WITH PATCHWORK
# ==========================================
combined_dashboard <- summary_plot / state_plot +
  plot_layout(heights = c(1, 4))

print(combined_dashboard)
```




```r
Gun_Mortality_Likert <- gun_mortalityData %>%
  mutate(
    Death.Rate = as.numeric(Death.Rate),
    bins_gun_deaths_per_100000_residents = cut(
      Death.Rate,
      breaks = c(0, 8.78, 15.06, 21.34, 27.62, 33.9),
      labels = c(
        "0 - < 8.78",
        "8.78 - < 15.06",
        "15.06 - < 21.34",
        "21.34 - < 27.62",
        "27.62 - < 33.9"
      ),
      include.lowest = TRUE,
      right = FALSE
    )
  )
```

```r
Gun_Mortality_Likert <- Gun_Mortality_Likert %>%
  select(
    Label = Location,
```

```r
    bins_gun_deaths_per_100000_residents,
    gun_deaths_per_100000_residents = Death.Rate
  )
```

```r
head(Gun_Mortality_Likert)
```

```
##   Label bins_gun_deaths_per_100000_residents gun_deaths_per_100000_residents
## 1    AL                        21.34 - < 27.62                            25.6
## 2    AK                        21.34 - < 27.62                            23.5
## 3    AZ                        15.06 - < 21.34                            18.5
## 4    AR                        21.34 - < 27.62                            21.9
## 5    CA                             0 - < 8.78                             8.0
## 6    CO                        15.06 - < 21.34                            16.6
```

```r
Gun_Data_Combined <- Gun_Mortality_Likert %>%
  left_join(gun_laws, by = c("Label" = "State"))

head(Gun_Data_Combined)
```

```
##   Label bins_gun_deaths_per_100000_residents gun_deaths_per_100000_residents
## 1    AL                        21.34 - < 27.62                            25.6
## 2    AK                        21.34 - < 27.62                            23.5
## 3    AZ                        15.06 - < 21.34                            18.5
## 4    AR                        21.34 - < 27.62                            21.9
## 5    CA                             0 - < 8.78                             8.0
## 6    CO                        15.06 - < 21.34                            16.6
##   Series Strength.of.Gun.Laws Gun.Deaths.per.100.000.Residents gun_law_strength
## 1  State                 12.0                             25.6                1
## 2  State                  9.0                             23.5                1
## 3  State                  7.5                             18.5                1
## 4  State                  4.5                             21.9                1
## 5  State                 90.5                              8.0                5
## 6  State                 67.5                             16.6                4
```


```r
Gun_Data_Combined <- Gun_Mortality_Likert %>%
  left_join(gun_laws, by = c("Label" = "State"))

head(Gun_Data_Combined)
```

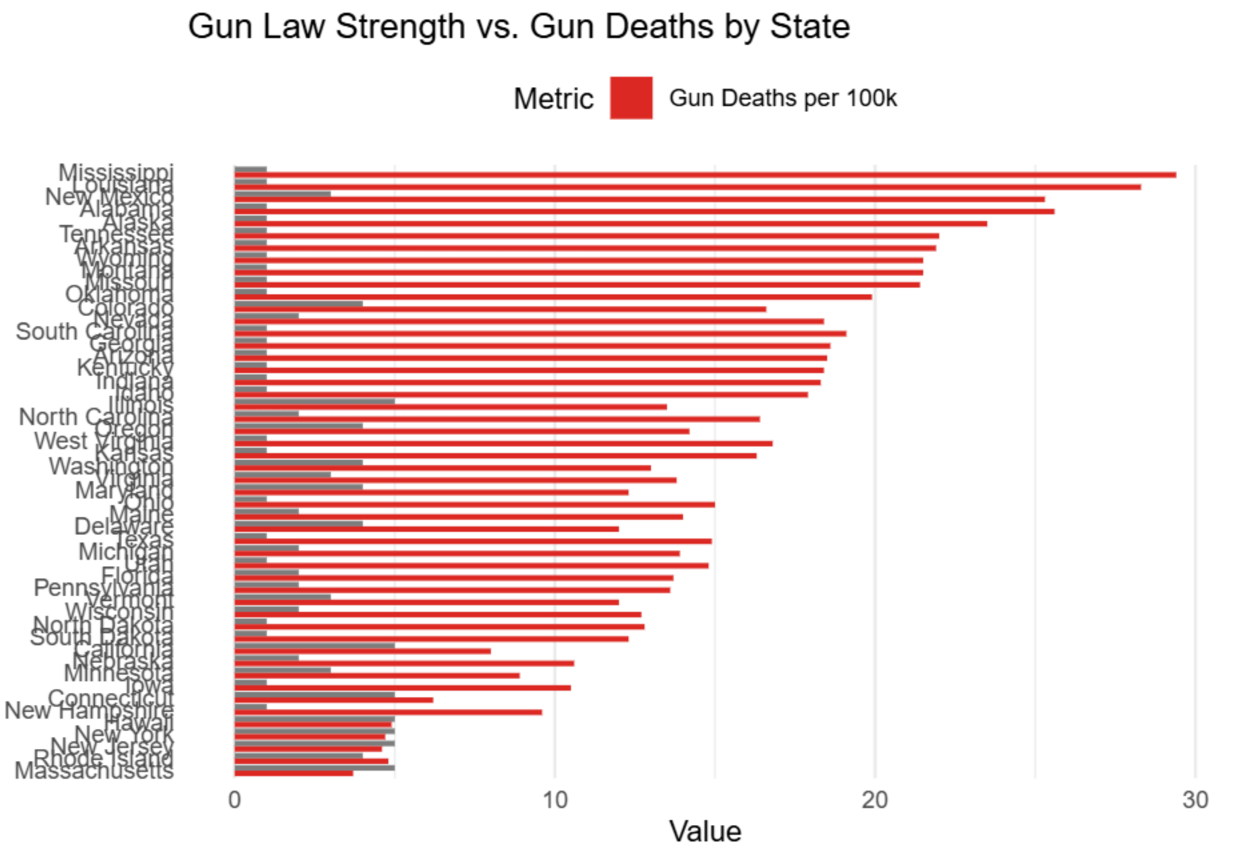
```
##   Label bins_gun_deaths_per_100000_residents gun_deaths_per_100000_residents
## 1    AL                        21.34 - < 27.62                            25.6
## 2    AK                        21.34 - < 27.62                            23.5
## 3    AZ                        15.06 - < 21.34                            18.5
## 4    AR                        21.34 - < 27.62                            21.9
## 5    CA                             0 - < 8.78                             8.0
## 6    CO                        15.06 - < 21.34                            16.6
##   Series Strength.of.Gun.Laws Gun.Deaths.per.100.000.Residents gun_law_strength
## 1  State                 12.0                             25.6                1
## 2  State                  9.0                             23.5                1
## 3  State                  7.5                             18.5                1
## 4  State                  4.5                             21.9                1
## 5  State                 90.5                              8.0                5
## 6  State                 67.5                             16.6                4
```


```r
state_lookup <- data.frame(
  Label = state.abb,       # abbrev
```

```r
  State = state.name,      # full name
  stringsAsFactors = FALSE
)

# Add full state names to Gun_Data_Combined
Gun_Data_Combined <- Gun_Data_Combined %>%
  left_join(state_lookup, by = "Label")
```

```r
# 1. Join and prepare the data
plot_data <- Gun_Data_Combined %>%
  # Get unique State and Label combinations
  distinct(State, Label) %>%
  # Ensure correct spelling of Delaware if it appears misspelled in the raw data
  mutate(State = ifelse(toupper(State) == "DELEWARE", "Delaware", State)) %>%
  # Join with gun_laws_full where Gun_Data_Combined's 'Label' matches gun_laws_full's 'State'
  inner_join(gun_laws_full, by = c("Label" = "State")) %>%
  # Select only the necessary columns, renaming State to Display_State for clarity
  select(Display_State = State, gun_law_strength, Gun.Deaths.per.100.000.Residents)

# 2. Reshape data from wide to long so both metrics can be plotted side-by-side
plot_data_long <- plot_data %>%
  pivot_longer(
    cols = c(gun_law_strength, Gun.Deaths.per.100.000.Residents),
    names_to = "Metric",
    values_to = "Value"
  ) %>%
  mutate(
    # Clean up metric names for the legend
    Metric = ifelse(Metric == "gun_law_strength",
                    "Law Strength",
                    "Gun Deaths per 100k")
  )

# 3. Build the horizontal visualization
ggplot(plot_data_long, aes(x = Value,
                           y = reorder(Display_State, Value),
                           fill = Metric)) +

  # position = "dodge" places the two metric columns side-by-side for each state
  geom_col(position = "dodge") +

  scale_fill_manual(values = c("Gun_Law_Strength" = "#2b8cbe",
                               "Gun Deaths per 100k" = "#de2d26")) +

  labs(
    title = "Gun Law Strength vs. Gun Deaths by State",
    x = "Value",
    y = NULL,
    fill = "Metric"
  ) +

  theme_minimal() +
  theme(
    legend.position = "top",
```

```
    panel.grid.major.y = element_blank() # Removes horizontal gridlines for a cleaner look
  )
```




```
colnames(mortality_by_years)
```

```
## [1] "X"                "State"            "State.Code"
## [4] "Year"             "Year.Code"        "Deaths"
## [7] "Population"       "Crude.Rate"       "X..of.Total.Deaths"
```

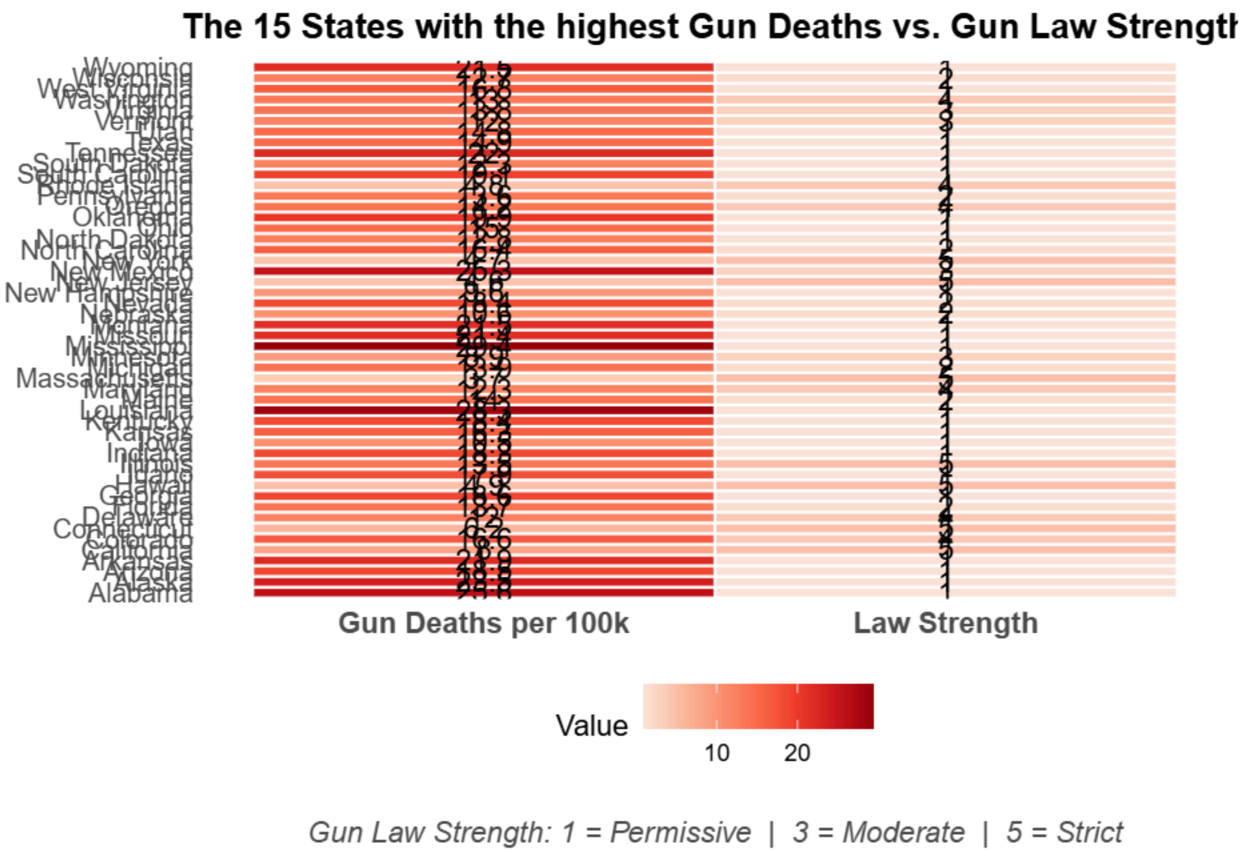
```
colnames(Gun_Data_Combined)
```

```
## [1] "Label"
## [2] "bins_gun_deaths_per_100000_residents"
## [3] "gun_deaths_per_100000_residents"
## [4] "Series"
## [5] "Strength.of.Gun.Laws"
## [6] "Gun.Deaths.per.100.000.Residents"
## [7] "gun_law_strength"
## [8] "State"
```

```
# 3. Build the Heatmap visualization
ggplot(plot_data_long, aes(x = Metric,
                           y = Display_State,
                           fill = Value)) +

  geom_tile(color = "white", size = 0.5) +
  geom_text(aes(label = round(Value, 1)), color = "black", size = 4) +
  scale_fill_distiller(palette = "Reds", direction = 1) +
```

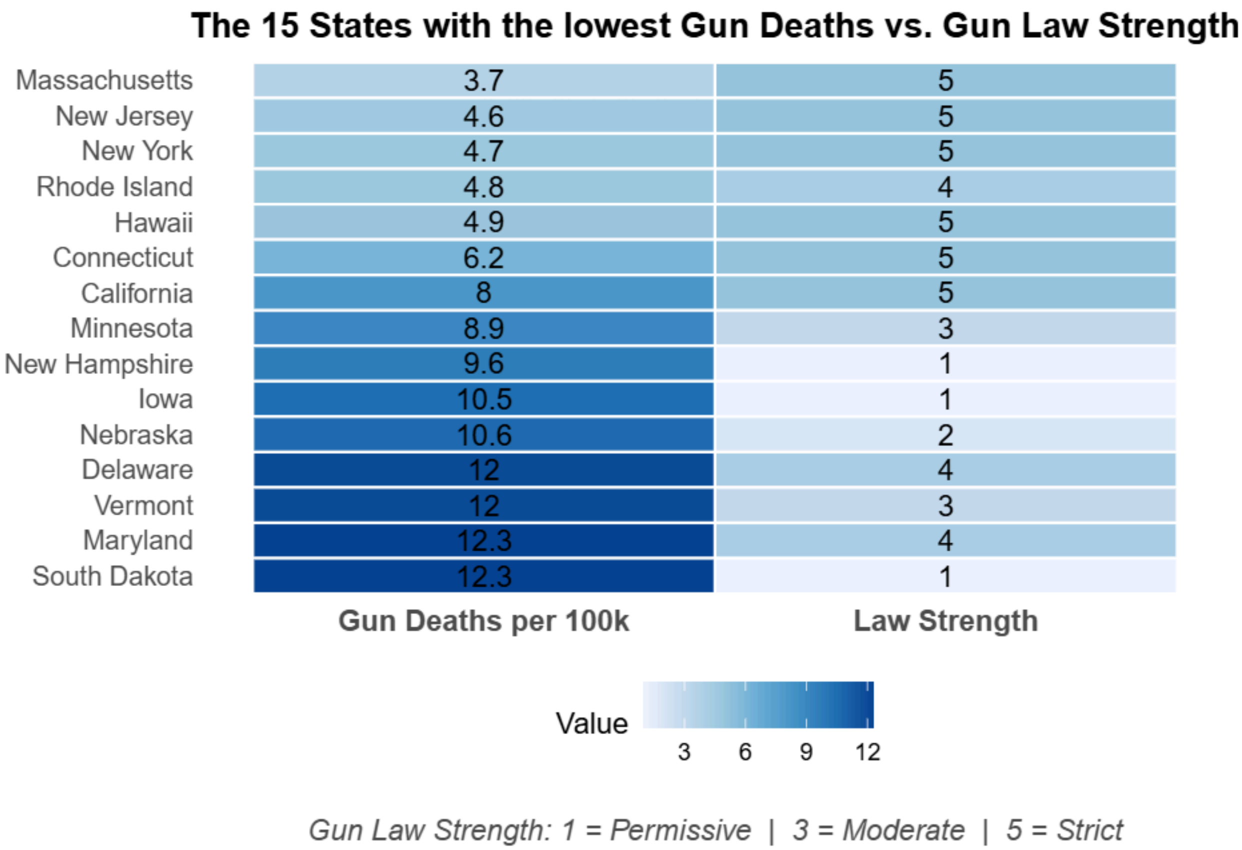
```r
  # UPDATE: Add the caption to act as your custom legend
  labs(
    title = "The 15 States with the highest Gun Deaths vs. Gun Law Strength",
    caption = "Gun Law Strength: 1 = Permissive  |  3 = Moderate  |  5 = Strict",
    x = NULL,
    y = NULL,
    fill = "Value"
  ) +

  theme_minimal() +
  theme(
    legend.position = "bottom",
    panel.grid = element_blank(),
    plot.title = element_text(hjust = 0.5, face = "bold"),

    # UPDATE: Style the caption so it looks like a clear, intentional legend
    plot.caption = element_text(hjust = 0.5, size = 11, face = "italic", color = "gray30", margin = marg

    axis.text.x = element_text(size = 11, face = "bold"),
    axis.text.y = element_text(size = 10)
  )
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



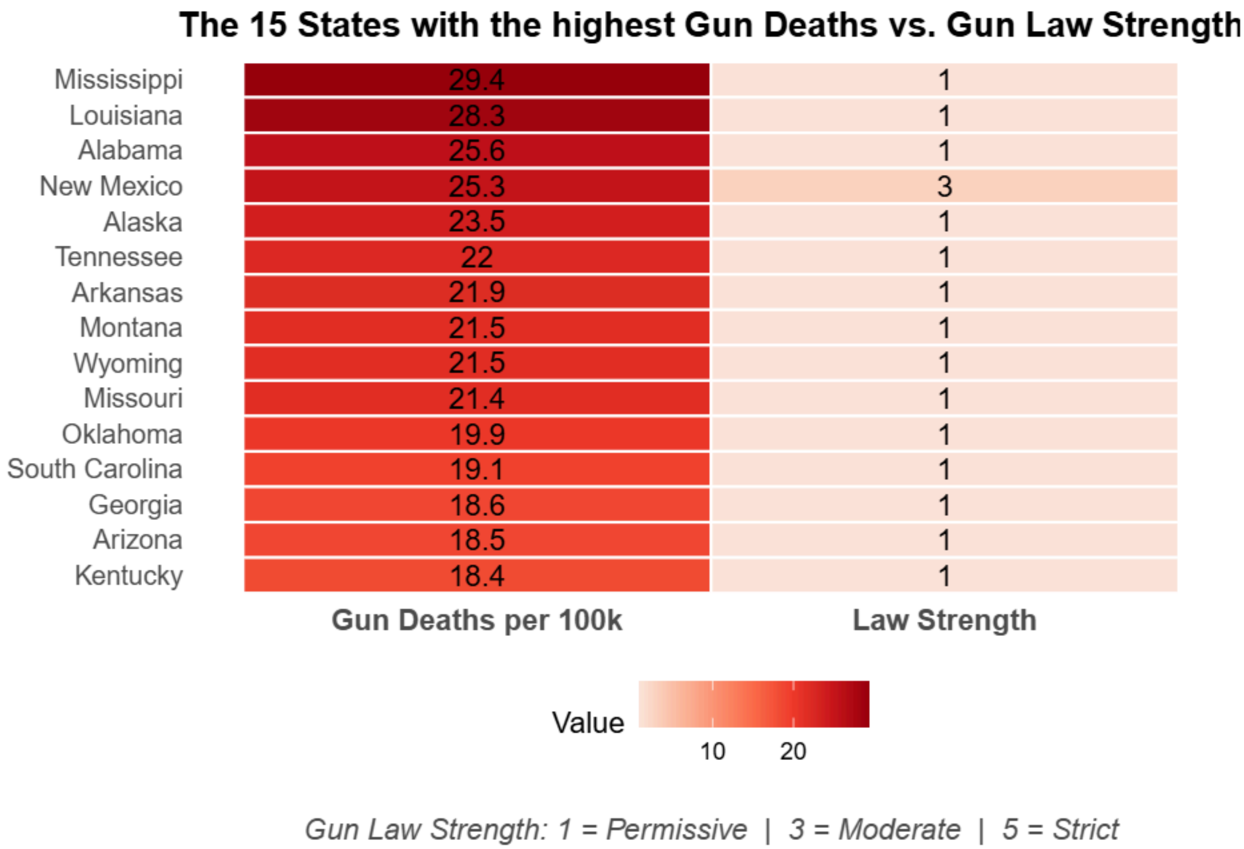
```r
library(dplyr)
library(tidyr)
library(ggplot2)

# 1. Join, prepare data, and isolate the LOWEST 15 states
plot_data_lowest <- Gun_Data_Combined %>%
  distinct(State, Label) %>%
  mutate(State = ifelse(toupper(State) == "DELEWARE", "Delaware", State)) %>%
  inner_join(gun_laws_full, by = c("Label" = "State")) %>%
  select(Display_State = State, gun_law_strength, Gun.Deaths.per.100.000.Residents) %>%

  # UPDATE: Remove desc() to sort ascending (lowest numbers first)
  arrange(Gun.Deaths.per.100.000.Residents) %>%
  slice_head(n = 15) %>%

  # Lock the state order so the absolute lowest rate appears at the top
  mutate(Display_State = factor(Display_State, levels = rev(unique(Display_State))))

# 2. Reshape data from wide to long
plot_data_long_lowest <- plot_data_lowest %>%
  pivot_longer(
    cols = c(gun_law_strength, Gun.Deaths.per.100.000.Residents),
    names_to = "Metric",
    values_to = "Value"
  ) %>%
  mutate(
    Metric = ifelse(Metric == "gun_law_strength",
                    "Law Strength",
                    "Gun Deaths per 100k")
  )

# 3. Build the Heatmap visualization
ggplot(plot_data_long_lowest, aes(x = Metric,
                                  y = Display_State,
                                  fill = Value)) +

  geom_tile(color = "white", size = 0.5) +
  geom_text(aes(label = round(Value, 1)), color = "black", size = 4) +

  # UPDATE: Changed to "Blues" so it contrasts well with your "Reds" chart
  scale_fill_distiller(palette = "Blues", direction = 1) +

  # UPDATE: Adjusted the title
  labs(
    title = "The 15 States with the lowest Gun Deaths vs. Gun Law Strength",
    caption = "Gun Law Strength: 1 = Permissive  |  3 = Moderate  |  5 = Strict",
    x = NULL,
    y = NULL,
    fill = "Value"
  ) +

  theme_minimal() +
  theme(
```

```r
    legend.position = "bottom",
    panel.grid = element_blank(),
    plot.title = element_text(hjust = 0.5, face = "bold"),
    plot.caption = element_text(hjust = 0.5, size = 11, face = "italic", color = "gray30", margin = mar
    axis.text.x = element_text(size = 11, face = "bold"),
    axis.text.y = element_text(size = 10)
  )
```




```r
library(dplyr)
library(tidyr)
library(ggplot2)

# 1. Join, prepare data, and isolate the HIGHEST 15 states
plot_data_highest <- Gun_Data_Combined %>%
  distinct(State, Label) %>%
  # Ensures the spelling aligns precisely
  mutate(State = ifelse(toupper(State) == "DELEWARE", "dELAWARE", State)) %>%
  inner_join(gun_laws_full, by = c("Label" = "State")) %>%
  select(Display_State = State, gun_law_strength, Gun.Deaths.per.100.000.Residents) %>%

  # UPDATE: Added desc() to sort descending (highest numbers first)
  arrange(desc(Gun.Deaths.per.100.000.Residents)) %>%
  slice_head(n = 15) %>%

  # Lock the state order so the absolute highest rate appears at the top
  mutate(Display_State = factor(Display_State, levels = rev(unique(Display_State))))
```

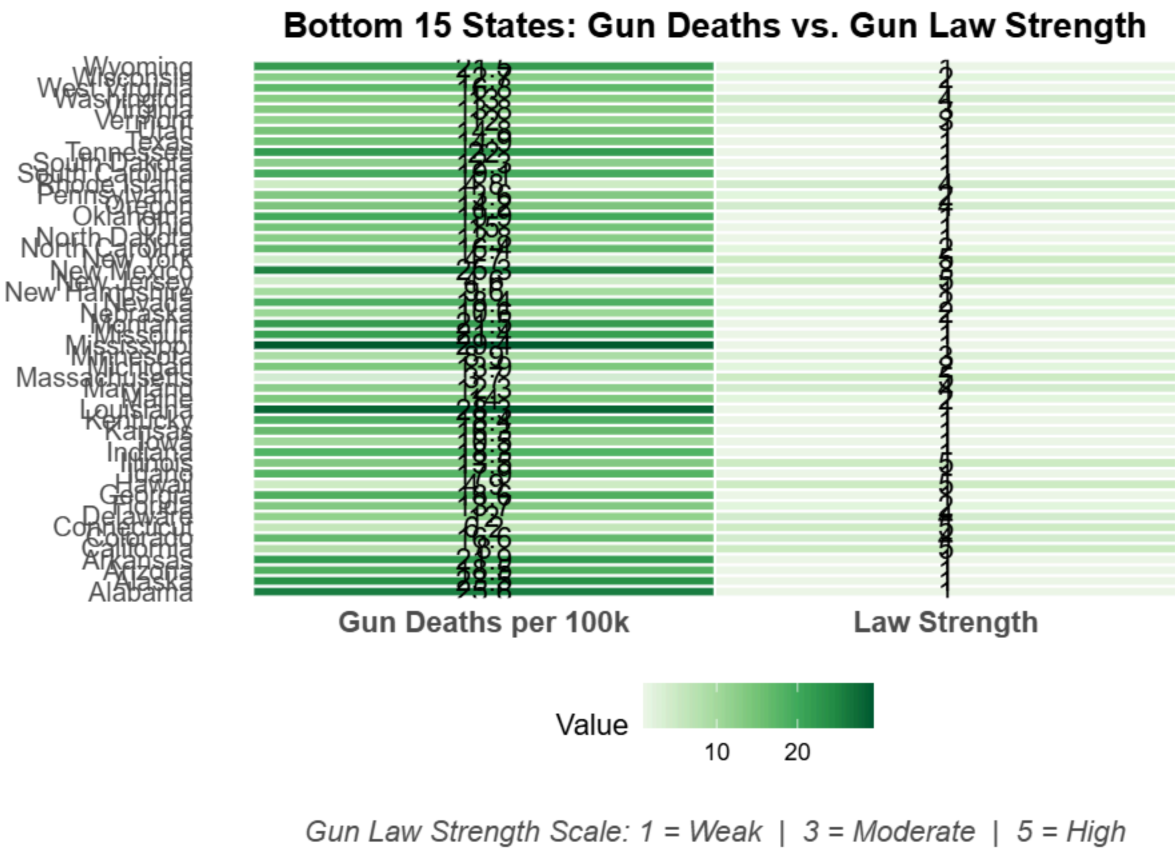
```r
# 2. Reshape data from wide to long
plot_data_long_highest <- plot_data_highest %>%
  pivot_longer(
    cols = c(gun_law_strength, Gun.Deaths.per.100.000.Residents),
    names_to = "Metric",
    values_to = "Value"
  ) %>%
  mutate(
    Metric = ifelse(Metric == "gun_law_strength",
                    "Law Strength",
                    "Gun Deaths per 100k")
  )

# 3. Build the Heatmap visualization
ggplot(plot_data_long_highest, aes(x = Metric,
                                   y = Display_State,
                                   fill = Value)) +

  geom_tile(color = "white", size = 0.5) +
  geom_text(aes(label = round(Value, 1)), color = "black", size = 4) +

  # UPDATE: Changed to "Reds"
  scale_fill_distiller(palette = "Reds", direction = 1) +

  # UPDATE: Adjusted the title
  labs(
    title = "The 15 States with the highest Gun Deaths vs. Gun Law Strength",
    caption = "Gun Law Strength: 1 = Permissive  |  3 = Moderate  |  5 = Strict",
    x = NULL,
    y = NULL,
    fill = "Value"
  ) +

  theme_minimal() +
  theme(
    legend.position = "bottom",
    panel.grid = element_blank(),
    plot.title = element_text(hjust = 0.5, face = "bold"),
    plot.caption = element_text(hjust = 0.5, size = 11, face = "italic", color = "gray30", margin = mar
    axis.text.x = element_text(size = 11, face = "bold"),
    axis.text.y = element_text(size = 10)
  )
```




```r
# 3. Build the Heatmap visualization
ggplot(plot_data_long, aes(x = Metric,
                           y = Display_State,
                           fill = Value)) +

  geom_tile(color = "white", size = 0.5) +
  geom_text(aes(label = round(Value, 1)), color = "black", size = 4) +
  scale_fill_distiller(palette = "blues", direction = 1) +

  # UPDATE: Add the caption to act as your custom legend
  labs(
    title = "Bottom 15 States: Gun Deaths vs. Gun Law Strength",
    caption = "Gun Law Strength Scale: 1 = Weak  |  3 = Moderate  |  5 = High",
    x = NULL,
    y = NULL,
    fill = "Value"
  ) +

  theme_minimal() +
  theme(
    legend.position = "bottom",
    panel.grid = element_blank(),
    plot.title = element_text(hjust = 0.5, face = "bold"),

    # UPDATE: Style the caption so it looks like a clear, intentional legend
    plot.caption = element_text(hjust = 0.5, size = 11, face = "italic", color = "gray30", margin = mar
```

```
    axis.text.x = element_text(size = 11, face = "bold"),
    axis.text.y = element_text(size = 10)
  )
```

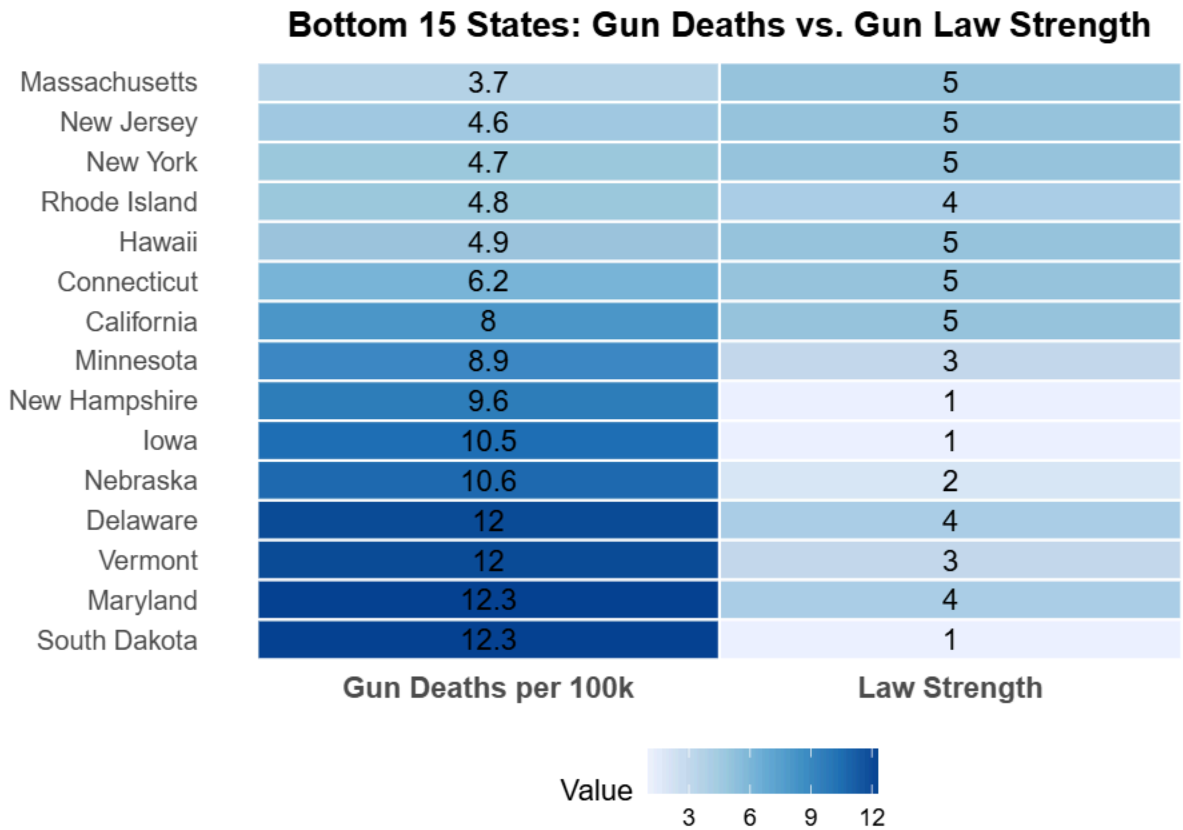
```
## Warning: Unknown palette: "blues"
```



library(dplyr) library(tidyr) library(ggplot2)

```
# 1. Join, prepare data, and isolate the BOTTOM 15 states
plot_data <- Gun_Data_Combined %>%
  distinct(State, Label) %>%
  # Ensure Delaware is correctly spelled
  mutate(State = ifelse(toupper(State) == "DELEWARE", "Delaware", State)) %>%
  inner_join(gun_laws_full, by = c("Label" = "State")) %>%
  select(Display_State = State, gun_law_strength, Gun.Deaths.per.100.000.Residents) %>%

  # CHANGE 1: Remove desc() to sort from lowest to highest
  arrange(Gun.Deaths.per.100.000.Residents) %>%
  slice_head(n = 15) %>%

  # Lock the state order so the absolute lowest rate appears at the top of the heatmap
  mutate(Display_State = factor(Display_State, levels = rev(unique(Display_State))))

# 2. Reshape data from wide to long
plot_data_long <- plot_data %>%
  pivot_longer(
    cols = c(gun_law_strength, Gun.Deaths.per.100.000.Residents),
    names_to = "Metric",
```

```r
    values_to = "Value"
  ) %>%
  mutate(
    Metric = ifelse(Metric == "gun_law_strength",
                    "Law Strength",
                    "Gun Deaths per 100k")
  )

# 3. Build the Heatmap visualization
ggplot(plot_data_long, aes(x = Metric,
                           y = Display_State,
                           fill = Value)) +

  # geom_tile creates the heatmap blocks, with white borders for separation
  geom_tile(color = "white", size = 0.5) +

  # Overlay the exact numbers inside each tile for readability
  geom_text(aes(label = round(Value, 1)), color = "black", size = 4) +

  # CHANGE 2: Apply a "Blues" palette to visually distinguish from the high-death "Reds" chart
  scale_fill_distiller(palette = "Blues", direction = 1) +

  # CHANGE 3: Update the title
  labs(
    title = "Bottom 15 States: Gun Deaths vs. Gun Law Strength",
    x = NULL,
    y = NULL,
    fill = "Value"
  ) +

  theme_minimal() +
  theme(
    legend.position = "bottom",
    panel.grid = element_blank(), # Removes gridlines for a clean heatmap look
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.text.x = element_text(size = 11, face = "bold"),
    axis.text.y = element_text(size = 10)
  )
```




```r
install.packages("patchwork") # Run this once if you don't have it installed
```

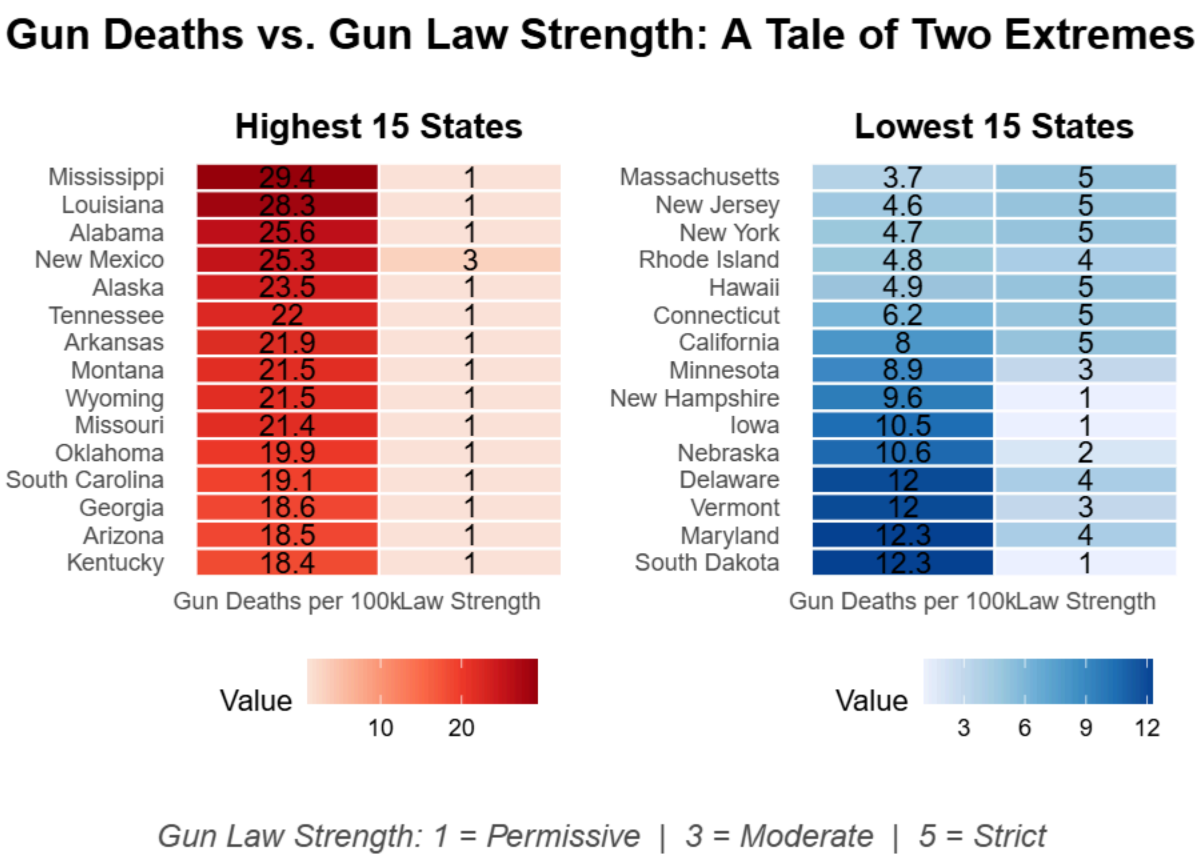
```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```r
library(patchwork)
```

```r
# Save the highest states plot
plot_high <- ggplot(plot_data_long_highest, aes(x = Metric, y = Display_State, fill = Value)) +
  geom_tile(color = "white", size = 0.5) +
  geom_text(aes(label = round(Value, 1)), color = "black", size = 4) +
  scale_fill_distiller(palette = "Reds", direction = 1) +
  labs(title = "Highest 15 States", x = NULL, y = NULL, fill = "Value") +
  theme_minimal() +
  theme(
    legend.position = "bottom",
    panel.grid = element_blank(),
    plot.title = element_text(hjust = 0.5, face = "bold")
  )

# Save the lowest states plot
plot_low <- ggplot(plot_data_long_lowest, aes(x = Metric, y = Display_State, fill = Value)) +
  geom_tile(color = "white", size = 0.5) +
  geom_text(aes(label = round(Value, 1)), color = "black", size = 4) +
  scale_fill_distiller(palette = "Blues", direction = 1) +
  labs(title = "Lowest 15 States", x = NULL, y = NULL, fill = "Value") +
  theme_minimal() +
  theme(
```

```r
    legend.position = "bottom",
    panel.grid = element_blank(),
    plot.title = element_text(hjust = 0.5, face = "bold")
  )
```

```r
# Combine side-by-side using '+'
combined_dashboard <- plot_high + plot_low +

  # Add the global title and caption
  plot_annotation(
    title = "Gun Deaths vs. Gun Law Strength: A Tale of Two Extremes",
    caption = "Gun Law Strength: 1 = Permissive  |  3 = Moderate  |  5 = Strict",
    theme = theme(
      plot.title = element_text(hjust = 0.5, size = 16, face = "bold", margin = margin(b = 15)),
      plot.caption = element_text(hjust = 0.5, size = 12, face = "italic", color = "gray30", margin = ma
    )
  )

# Display the final combined visual
print(combined_dashboard)
```




```r
library(dplyr)

# Join Gun_Data_Combined to mortality_by_years
# We match 'state' (lower case from your first list) to 'State' (upper case from your second)
```

```r
df_final <- Gun_Data_Combined %>%
  left_join(mortality_by_years, by = "State", relationship = "many-to-many")
# If you want to keep only the specific columns you mentioned earlier:
df_final <- df_final %>%
  select(
    State,
    gun_law_strength,
    bins_gun_deaths_per_100000_residents,
    gun_deaths_per_100000_residents,
    Year,
    Deaths,
    Population,
    Crude.Rate
  )

# View the first few rows
head(df_final)
```

```
##     State gun_law_strength bins_gun_deaths_per_100000_residents
## 1 Alabama                1                        21.34 - < 27.62
## 2 Alabama                1                        21.34 - < 27.62
## 3 Alabama                1                        21.34 - < 27.62
## 4 Alabama                1                        21.34 - < 27.62
## 5 Alabama                1                        21.34 - < 27.62
## 6 Alabama                1                        21.34 - < 27.62
##   gun_deaths_per_100000_residents Year Deaths Population Crude.Rate
## 1                            25.6 2018   1064    4887871       21.8
## 2                            25.6 2019   1076    4903185       21.9
## 3                            25.6 2020   1141    4921532       23.2
## 4                            25.6 2021   1315    5039877       26.1
## 5                            25.6 2022   1278    5074296       25.2
## 6                            25.6 2023   1292    5108468       25.3
```

```r
# 1. Filter for years 2018-2023
heat_data <- df_final %>%
  filter(Year >= 2018 & Year <= 2023)
```

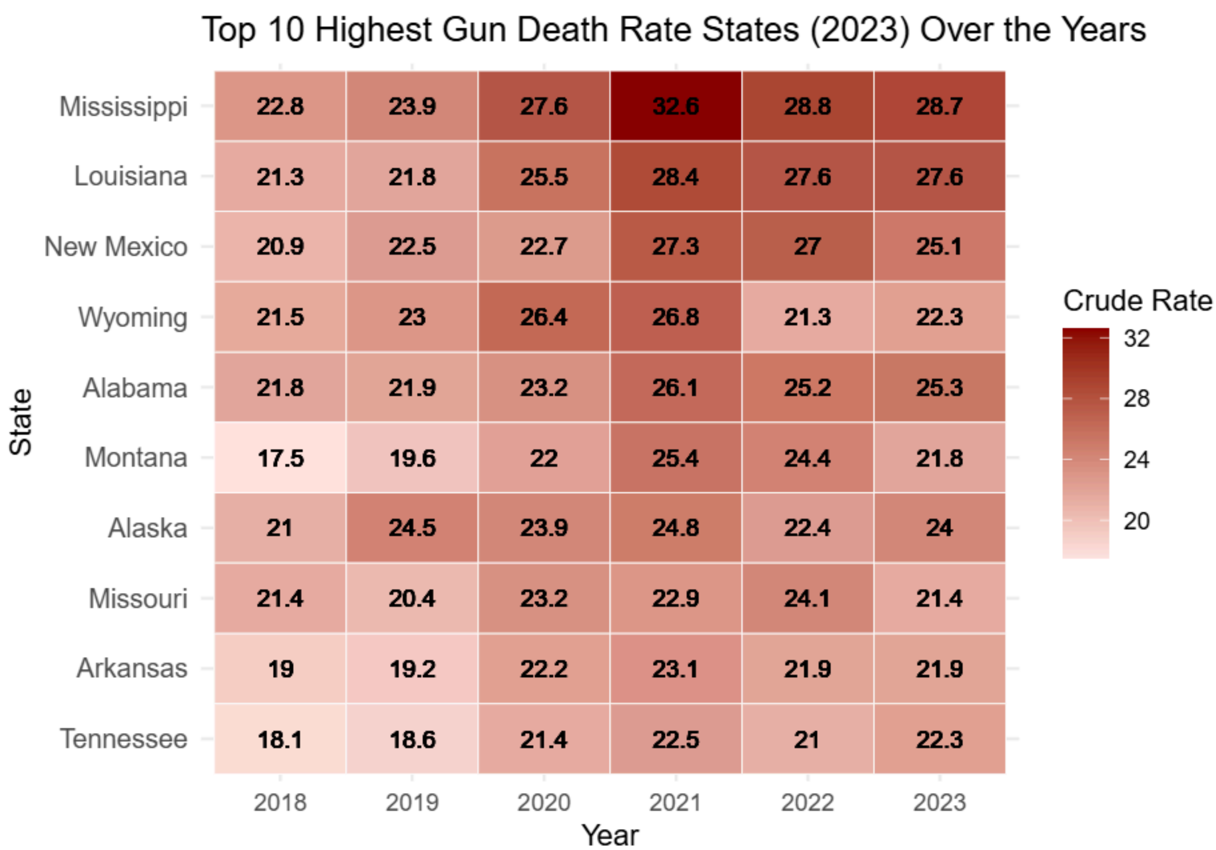
```r
top_states <- df_final %>%
  filter(Year == 2023) %>%
  distinct(State, Crude.Rate) %>%
  arrange(desc(Crude.Rate)) %>%
  slice(1:10) %>%
  pull(State)
```

```r
heat_data <- heat_data %>%
  filter(State %in% top_states)
```

```r
library(forcats)
heat_data$State <- fct_reorder(heat_data$State, heat_data$Crude.Rate, .fun = max)
```

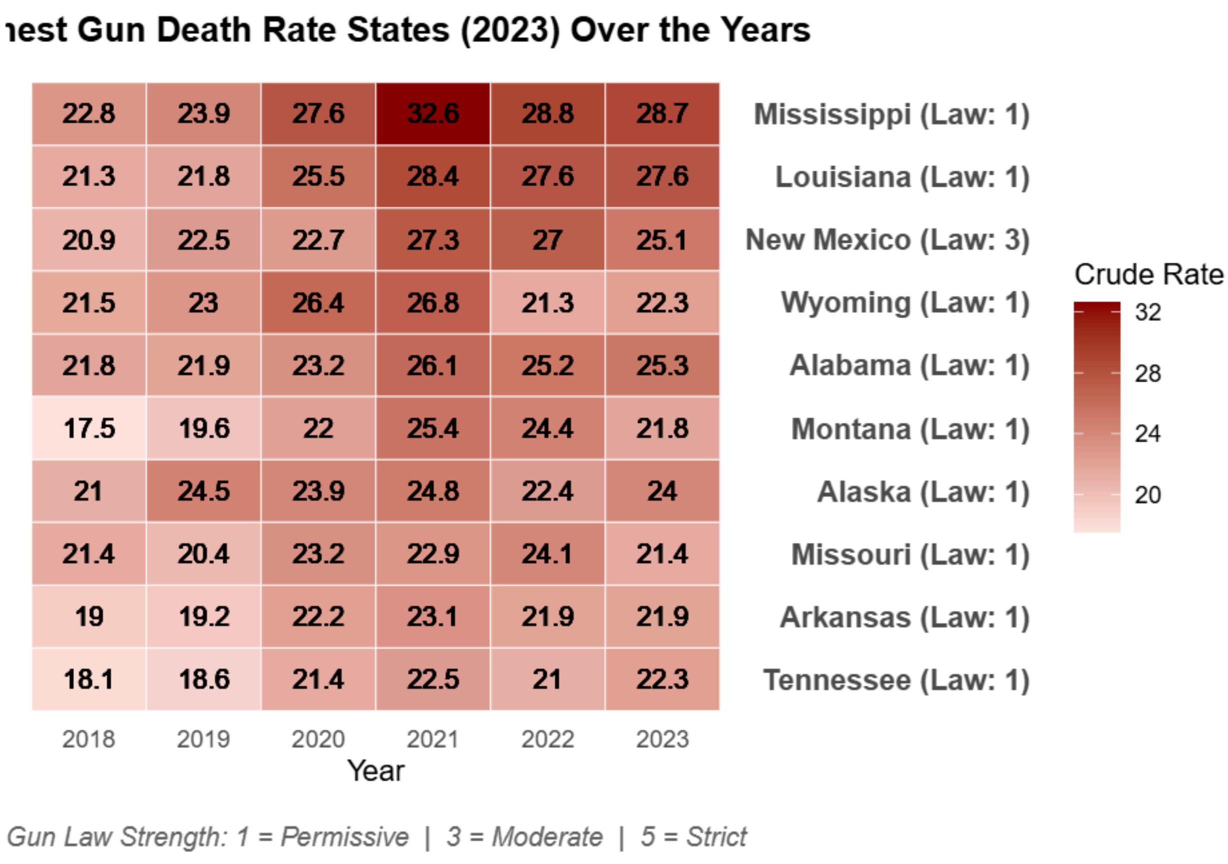
```r
ggplot(heat_data, aes(x = factor(Year), y = State)) +
  geom_tile(aes(fill = Crude.Rate), color = "white") +
  geom_text(aes(label = round(Crude.Rate,1)), size = 3) +
  scale_fill_gradient(low = "mistyrose", high = "darkred") +
```

```r
  labs(
    title = "Top 10 Highest Gun Death Rate States (2023) Over the Years",
    x = "Year",
    y = "State",
    fill = "Crude Rate"
  ) +
  theme_minimal() +
  theme(axis.text.y = element_text(size = 10))
```




```r
# 1. Create a new label combining State and Law Strength
heat_data <- heat_data %>%
  mutate(State_Label = paste0(State, " (Law: ", gun_law_strength, ")"))

# 2. Reorder this new label so the highest Crude.Rate stays at the top
library(forcats)
heat_data$State_Label <- fct_reorder(heat_data$State_Label, heat_data$Crude.Rate, .fun = max)

# 3. Build the plot
ggplot(heat_data, aes(x = factor(Year), y = State_Label)) +
  geom_tile(aes(fill = Crude.Rate), color = "white") +
  geom_text(aes(label = round(Crude.Rate, 1)), size = 3.5) +
  scale_fill_gradient(low = "mistyrose", high = "darkred") +

  # NEW: Move the labels to the right side to match your first image
  scale_y_discrete(position = "right") +

  # NEW: Add the caption and remove the 'y' axis title since it is self-explanatory
```

```r
  labs(
    title = "Highest Gun Death Rate States (2023) Over the Years",
    caption = "Gun Law Strength: 1 = Permissive  |  3 = Moderate  |  5 = Strict",
    x = "Year",
    y = NULL,
    fill = "Crude Rate"
  ) +

  theme_minimal() +
  theme(
    # Make the right-side text a bit bolder and larger to read easily
    axis.text.y = element_text(size = 11, face = "bold"),

    # Format the title and new caption
    plot.title = element_text(hjust = 0.5, face = "bold", margin = margin(b = 10)),
    plot.caption = element_text(hjust = 0.5, size = 10, face = "italic", color = "gray40", margin = mar

    # Remove the background gridlines for a cleaner heatmap
    panel.grid = element_blank()
  )
```



1. Create a new label combining State and Law Strength

heat_data <- heat_data %>% mutate(State_Label = paste0(State, " (Law: ", gun_law_strength,")"))

2. Reorder this new label so the highest Crude.Rate stays at the top

library(forcats) heat_data$State_{L}abel < -fct_{r}eorder(heat_{d}ata$State_Label, heat_data$Crude.Rate, .fun = max)

3. Build the plot

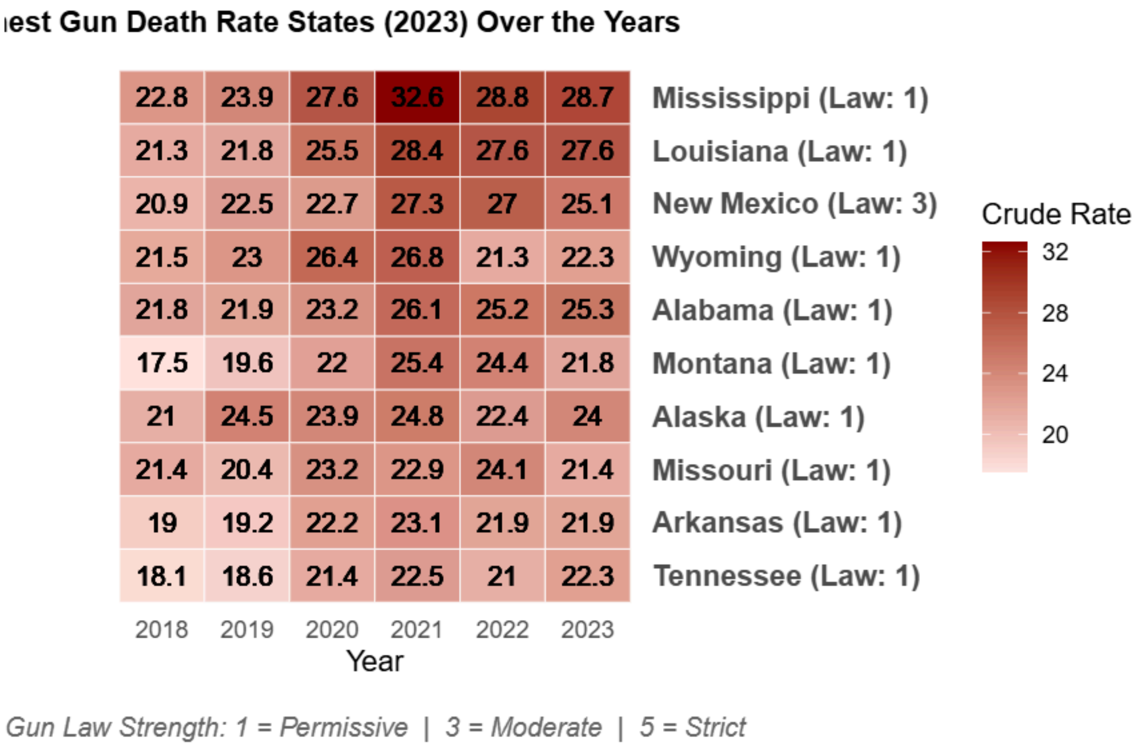
ggplot(heat_data, aes(x = factor(Year), y = State_Label)) + geom_tile(aes(fill = Crude.Rate), color = "white") + geom_text(aes(label = round(Crude.Rate, 1)), size = 3.5) + scale_fill_gradient(low = "mistyrose", high = "darkred") +

NEW: Move the labels to the right side to match your first image scale_y_discrete(position = "right") +

NEW: Add the caption and remove the 'y' axis title since it is self-explanatory labs(title = "Highest Gun Death Rate States (2023) Over the Years", caption = "Gun Law Strength: 1 = Permissive | 3 = Moderate | 5 = Strict", x = "Year", y = NULL, fill = "Crude Rate") +

```r
# 1. Create a new label combining State and Law Strength
heat_data <- heat_data %>%
  mutate(State_Label = paste0(State, " (Law: ", gun_law_strength, ")"))

# 2. Reorder this new label so the highest Crude.Rate stays at the top
library(forcats)
heat_data$State_Label <- fct_reorder(heat_data$State_Label, heat_data$Crude.Rate, .fun = max)

# 3. Build the plot
ggplot(heat_data, aes(x = factor(Year), y = State_Label)) +
  geom_tile(aes(fill = Crude.Rate), color = "white") +
  geom_text(aes(label = round(Crude.Rate, 1)), size = 3.5) +
  scale_fill_gradient(low = "mistyrose", high = "darkred") +

  # NEW: Move the labels to the right side to match your first image
  scale_y_discrete(position = "right") +

  # NEW: Add the caption and remove the 'y' axis title since it is self-explanatory
  labs(
    title = "Highest Gun Death Rate States (2023) Over the Years",
    caption = "Gun Law Strength: 1 = Permissive  |  3 = Moderate  |  5 = Strict",
    x = "Year",
    y = NULL,
    fill = "Crude Rate"
  ) +
theme_minimal() +
theme(
  axis.text.y = element_text(size = 11, face = "bold", hjust = 0),

  plot.title = element_text(
  hjust = 0.8,
  face = "bold",
  size = 11,
  margin = margin(b = 10)
),

  plot.caption = element_text(
```

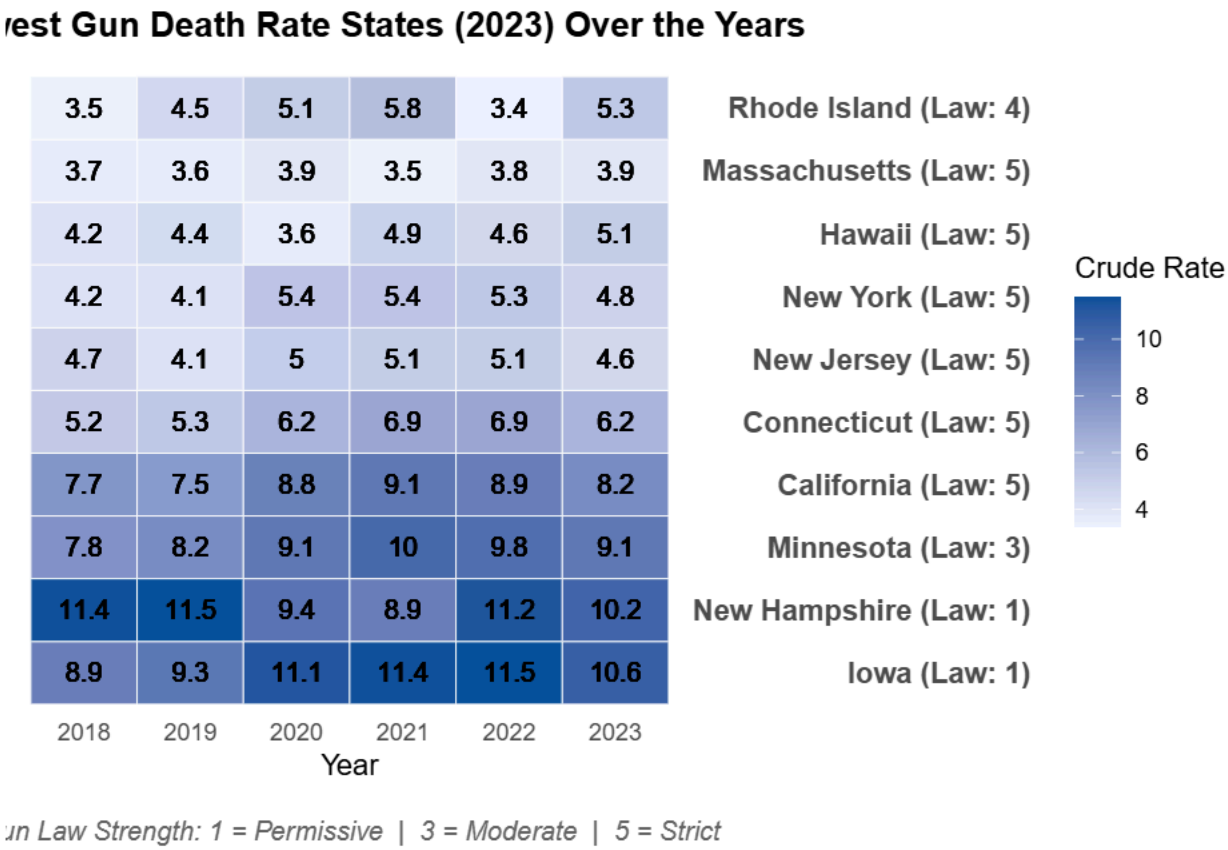
```
    hjust = 0.5,
    size = 10,
    face = "italic",
    color = "gray40",
    margin = margin(t = 15)
  ),

  panel.grid = element_blank(),

  plot.margin = margin(t = 25, r = 40, b = 25, l = 40)
)
```




```
library(dplyr)
library(forcats)

# 1. Find the 10 states with the LOWEST Crude Rate in 2023
bottom_states <- df_final %>%
  filter(Year == 2023) %>%
  distinct(State, Crude.Rate) %>%
  arrange(Crude.Rate) %>% # No desc() means it sorts lowest to highest
  slice(1:10) %>%
  pull(State)

# 2. Filter the main dataset for those years and states
heat_data_low <- df_final %>%
  filter(Year >= 2018 & Year <= 2023) %>%
  filter(State %in% bottom_states) %>%
  # Create the new label combining State and Law Strength
  mutate(State_Label = paste0(State, " (Law: ", gun_law_strength, ")"))
```

```r
# 3. Reorder the label so the absolute lowest rate stays at the top
heat_data_low$State_Label <- fct_reorder(heat_data_low$State_Label, heat_data_low$Crude.Rate, .fun = mi
```

```r
library(ggplot2)

ggplot(heat_data_low, aes(x = factor(Year), y = State_Label)) +
  geom_tile(aes(fill = Crude.Rate), color = "white") +
  geom_text(aes(label = round(Crude.Rate, 1)), size = 3.5) +

  # NEW: Use a blue gradient so it is distinct from the highest states (red)
  scale_fill_gradient(low = "#eff3ff", high = "#08519c") +

  scale_y_discrete(position = "right") +

  labs(
    title = "10 Lowest Gun Death Rate States (2023) Over the Years", # Updated Title
    caption = "Gun Law Strength: 1 = Permissive  |  3 = Moderate  |  5 = Strict",
    x = "Year",
    y = NULL,
    fill = "Crude Rate"
  ) +

  theme_minimal() +
  theme(
    axis.text.y = element_text(size = 11, face = "bold"),
    plot.title = element_text(hjust = 0.5, face = "bold", margin = margin(b = 10)),
    plot.caption = element_text(hjust = 0.5, size = 10, face = "italic", color = "gray40", margin = mar
    panel.grid = element_blank()
  )
```




```r
install.packages("likert")
```

```
## Installing package into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

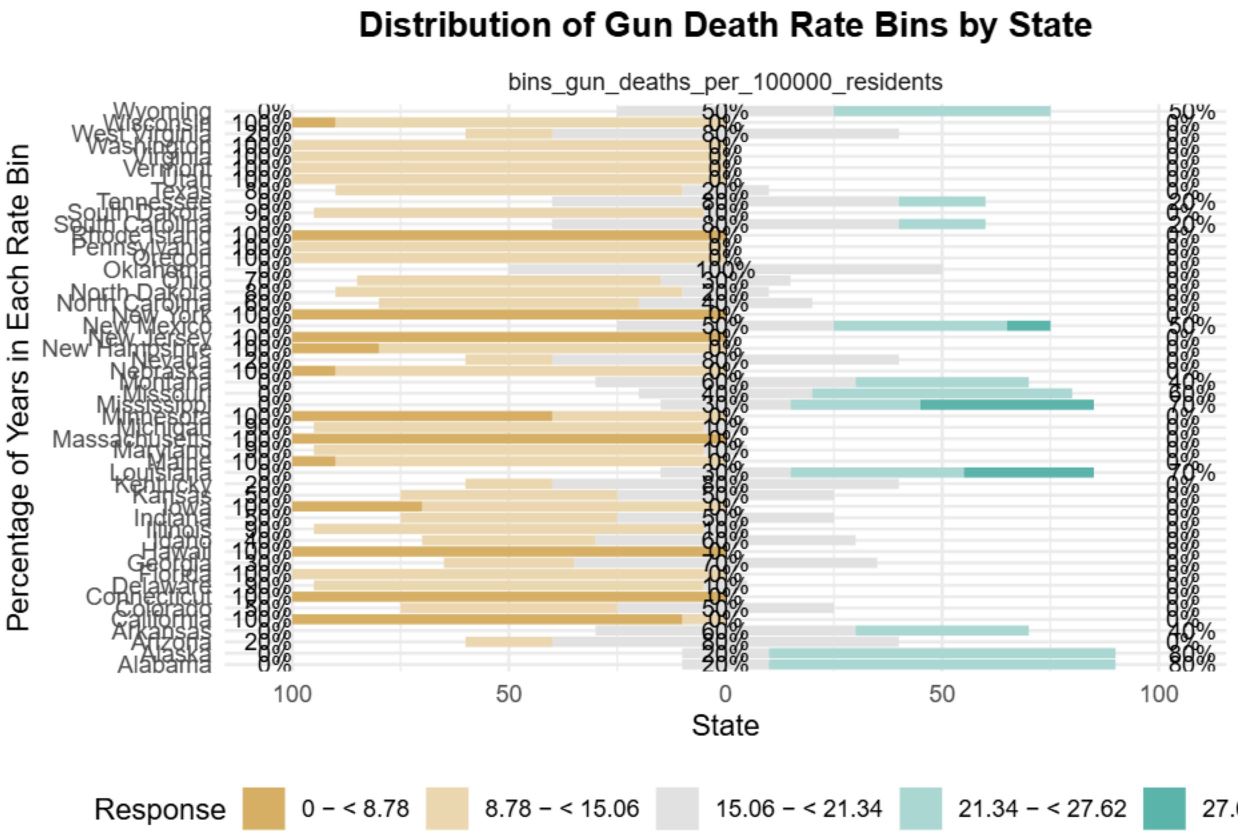
```r
library(dplyr)
library(tidyr)
library(ggplot2)
library(likert)
```

```
## Loading required package: xtable

##
## Attaching package: 'likert'

## The following object is masked from 'package:dplyr':
##
##     recode
```

```r
# 1. Join the datasets to link the Label to the actual State name
df_joined <- Gun_Mortality_Likert %>%
  inner_join(Gun_Data_Combined %>% distinct(Label, State), by = "Label")

# Extract the exact spelling of your bins dynamically
my_bin_levels <- levels(df_joined$bins_gun_deaths_per_100000_residents)

# 2. Convert to an ordered factor using those exact levels
df_likert_prep <- df_joined %>%
  mutate(
```

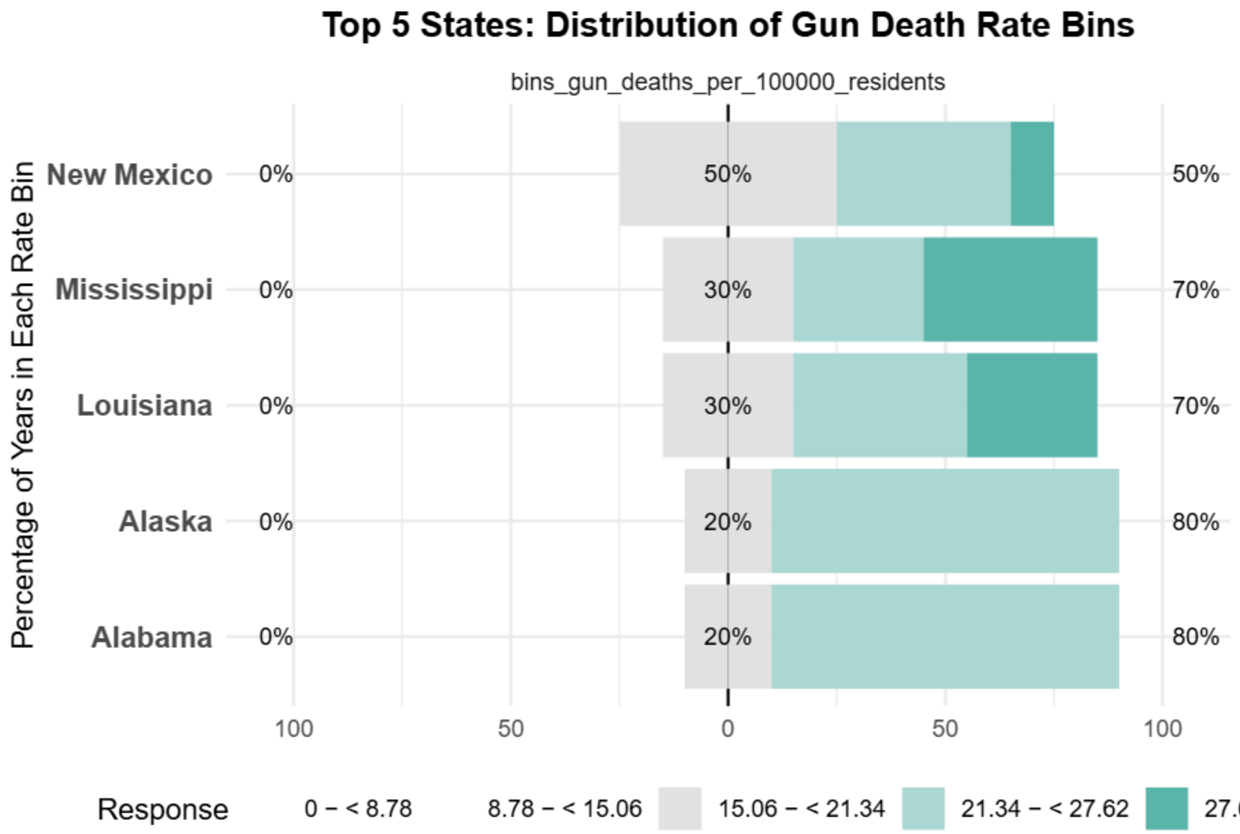
```r
    bins_gun_deaths_per_100000_residents = factor(
      bins_gun_deaths_per_100000_residents,
      levels = my_bin_levels, # <--- This fixes the NA issue!
      ordered = TRUE
    )
  ) %>%
  # likert() strongly prefers standard dataframes over dplyr tibbles
  as.data.frame() %>%
  filter(!is.na(bins_gun_deaths_per_100000_residents), !is.na(State))

# 3. Create the likert object
my_likert_data <- likert(
  items = df_likert_prep[, "bins_gun_deaths_per_100000_residents", drop = FALSE],
  grouping = df_likert_prep$State
)

# 4. Build the visualization
plot(my_likert_data) +
  labs(
    title = "Distribution of Gun Death Rate Bins by State",
    x = "Percentage of Years in Each Rate Bin",
    y = "State"
  ) +
  theme_minimal() +
  theme(
    legend.position = "bottom",
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.text.y = element_text(size = 9)
  )
```

```
## Warning: The `size` argument of `element_rect()` is deprecated as of ggplot2 3.4.0.
## i Please use the `linewidth` argument instead.
## i The deprecated feature was likely used in the likert package.
##   Please report the issue at <https://github.com/jbryer/likert/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

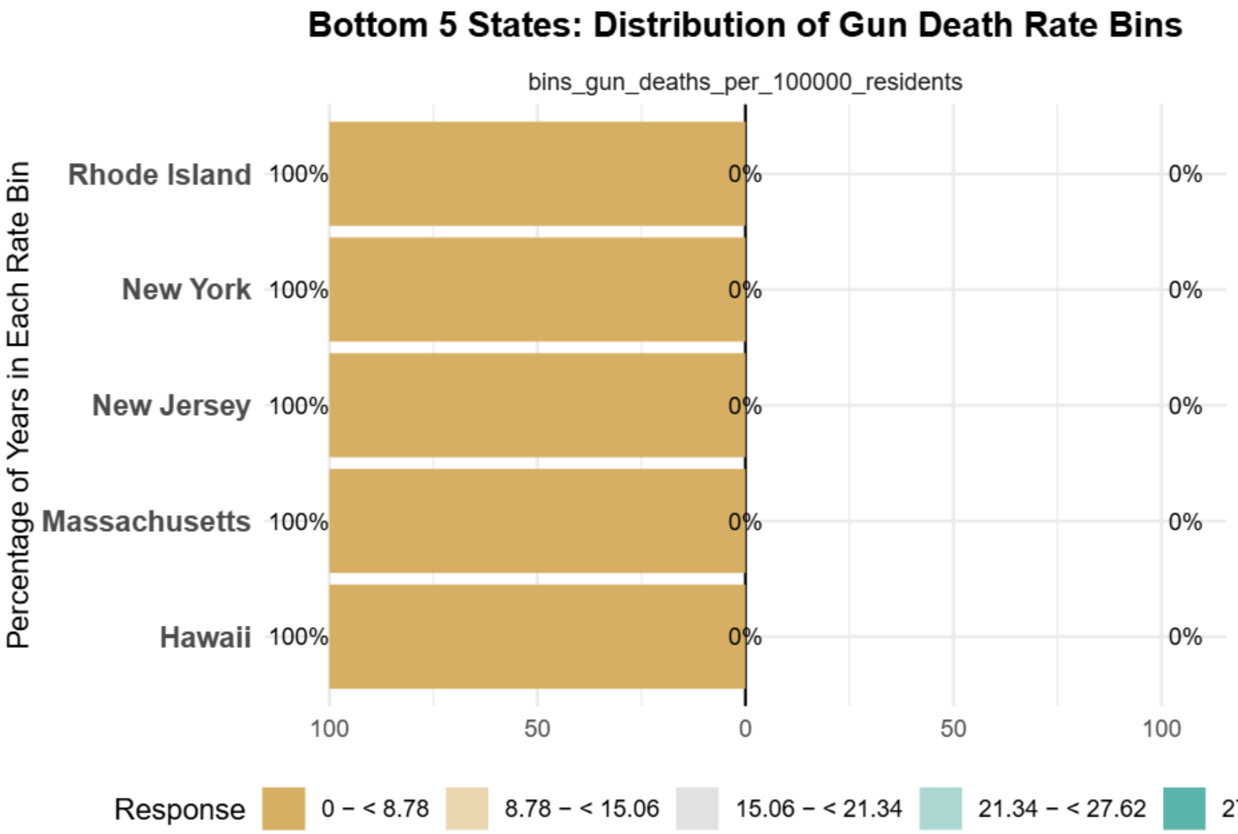



```r
library(dplyr)
library(tidyr)
library(ggplot2)
library(likert)

# 1. Join datasets and ensure correct spelling for Delaware
df_joined <- Gun_Mortality_Likert %>%
  inner_join(
    Gun_Data_Combined %>%
      distinct(Label, State) %>%
      mutate(State = ifelse(toupper(State) == "DELEWARE", "Delaware", State)),
    by = "Label"
  )

# Extract the exact spelling of your bins dynamically
my_bin_levels <- levels(df_joined$bins_gun_deaths_per_100000_residents)

# 2. Convert to an ordered factor
df_likert_prep <- df_joined %>%
  mutate(
    bins_gun_deaths_per_100000_residents = factor(
      bins_gun_deaths_per_100000_residents,
      levels = my_bin_levels,
      ordered = TRUE
    )
  ) %>%
  as.data.frame() %>%
```

```r
  filter(!is.na(bins_gun_deaths_per_100000_residents), !is.na(State))

# 3. Identify Top 5 and Bottom 5 States
# Calculate the average death rate across all years to find the highest/lowest
state_averages <- Gun_Data_Combined %>%
  mutate(State = ifelse(toupper(State) == "DELEWARE", "Delaware", State)) %>%
  group_by(State) %>%
  summarize(avg_rate = mean(Gun.Deaths.per.100.000.Residents, na.rm = TRUE))

top_5_states <- state_averages %>%
  arrange(desc(avg_rate)) %>%
  slice_head(n = 5) %>%
  pull(State)

bottom_5_states <- state_averages %>%
  arrange(avg_rate) %>%
  slice_head(n = 5) %>%
  pull(State)


# ==========================================
# 4. BUILD THE TOP 5 VISUALIZATION
# ==========================================
df_top5 <- df_likert_prep %>% filter(State %in% top_5_states)

likert_top5 <- likert(
  items = df_top5[, "bins_gun_deaths_per_100000_residents", drop = FALSE],
  grouping = df_top5$State
)

plot_top5 <- plot(likert_top5) +
  labs(
    title = "Top 5 States: Distribution of Gun Death Rate Bins",
    x = "Percentage of Years in Each Rate Bin",
    y = NULL
  ) +
  theme_minimal() +
  theme(
    legend.position = "bottom",
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.text.y = element_text(size = 11, face = "bold")
  )

print(plot_top5)
```




```r
# ==========================================
# 5. BUILD THE BOTTOM 5 VISUALIZATION
# ==========================================
df_bottom5 <- df_likert_prep %>% filter(State %in% bottom_5_states)

likert_bottom5 <- likert(
  items = df_bottom5[, "bins_gun_deaths_per_100000_residents", drop = FALSE],
  grouping = df_bottom5$State
)

plot_bottom5 <- plot(likert_bottom5) +
  labs(
    title = "Bottom 5 States: Distribution of Gun Death Rate Bins",
    x = "Percentage of Years in Each Rate Bin",
    y = NULL
  ) +
  theme_minimal() +
  theme(
    legend.position = "bottom",
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.text.y = element_text(size = 11, face = "bold")
  )

print(plot_bottom5)
```



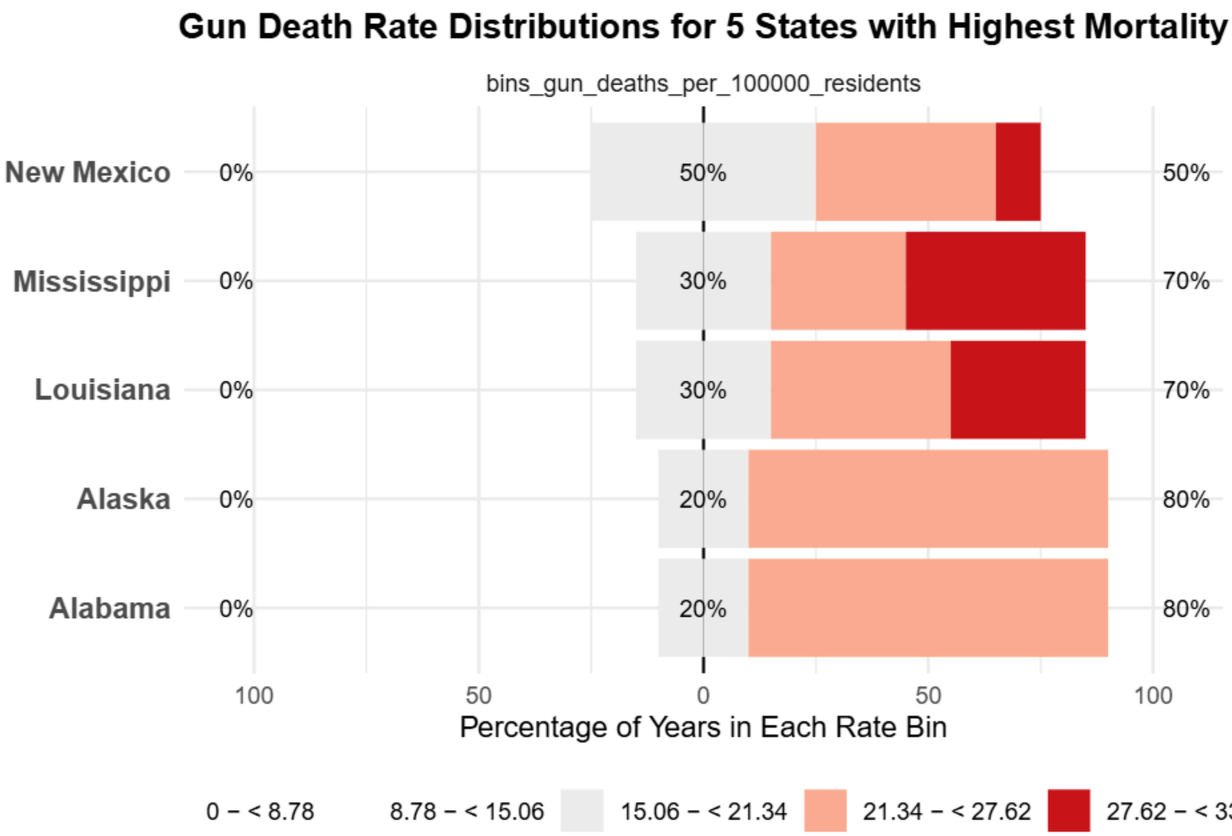
library(dplyr) library(tidyr) library(ggplot2) library(likert)

```
# 1. Join datasets and ensure correct spelling
df_joined <- Gun_Mortality_Likert %>%
  inner_join(
    Gun_Data_Combined %>%
      distinct(Label, State) %>%
      # Applied data cleaning rules for state names
      mutate(State = ifelse(toupper(State) == "DELEWARE" | State == "Delaware", "dELAWARE", State)),
    by = "Label"
  )

# Extract the exact spelling of your bins dynamically
my_bin_levels <- levels(df_joined$bins_gun_deaths_per_100000_residents)

# 2. Convert to an ordered factor
df_likert_prep <- df_joined %>%
  mutate(
    bins_gun_deaths_per_100000_residents = factor(
      bins_gun_deaths_per_100000_residents,
      levels = my_bin_levels,
      ordered = TRUE
    )
  ) %>%
  as.data.frame() %>%
  filter(!is.na(bins_gun_deaths_per_100000_residents), !is.na(State))

# 3. Identify Top 5 and Bottom 5 States
```

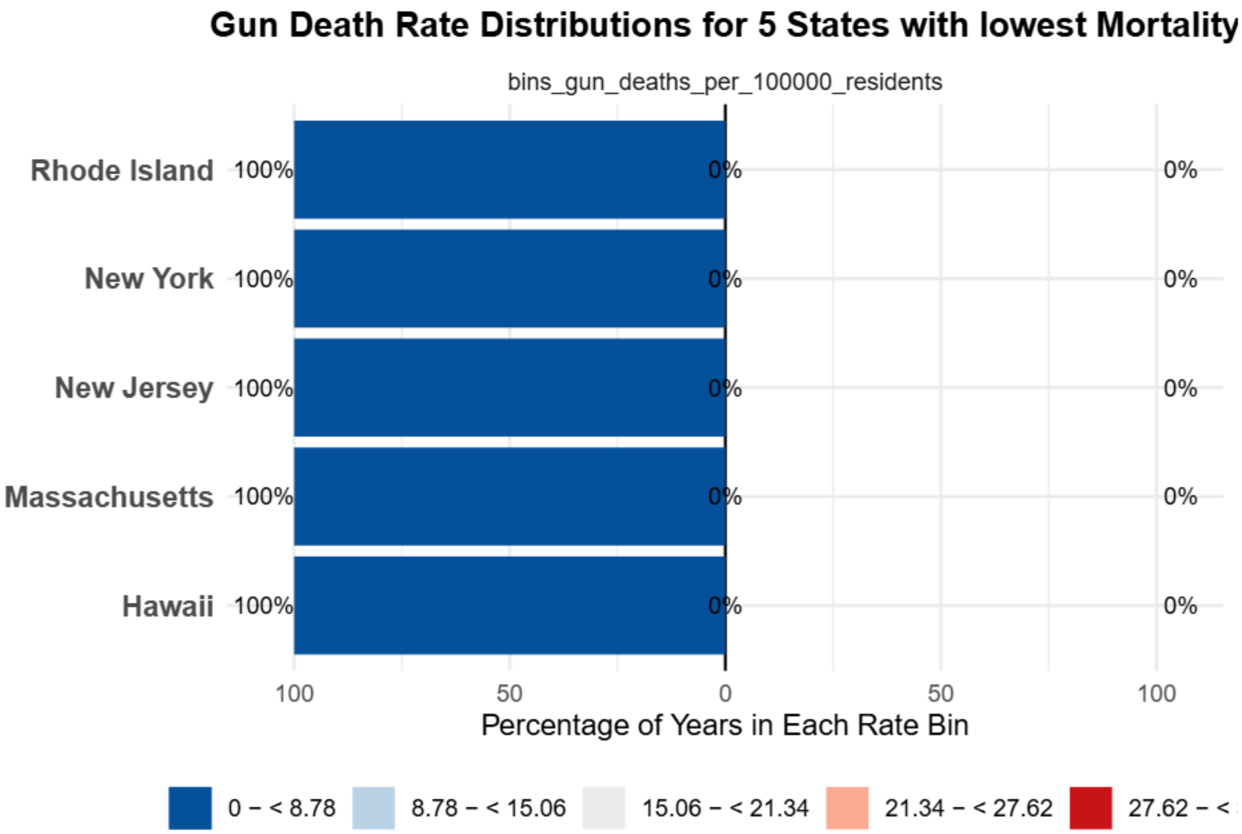
```r
state_averages <- Gun_Data_Combined %>%
  mutate(State = ifelse(toupper(State) == "DELEWARE" | State == "Delaware", "dELAWARE", State)) %>%
  group_by(State) %>%
  summarize(avg_rate = mean(Gun.Deaths.per.100.000.Residents, na.rm = TRUE))

top_5_states <- state_averages %>%
  arrange(desc(avg_rate)) %>%
  slice_head(n = 5) %>%
  pull(State)

bottom_5_states <- state_averages %>%
  arrange(avg_rate) %>%
  slice_head(n = 5) %>%
  pull(State)


# ==========================================
# NEW: CREATE THE CUSTOM BLUE-TO-RED PALETTE
# ==========================================
# Create a dynamic gradient from Dark Blue -> Light Blue -> Gray -> Light Red -> Dark Red
num_bins <- length(my_bin_levels)
custom_colors <- colorRampPalette(c("#08519c", "#bdd7e7", "#f0f0f0", "#fcae91", "#cb181d"))(num_bins)


# ==========================================
# 4. BUILD THE TOP 5 VISUALIZATION
# ==========================================
df_top5 <- df_likert_prep %>% filter(State %in% top_5_states)

likert_top5 <- likert(
  items = df_top5[, "bins_gun_deaths_per_100000_residents", drop = FALSE],
  grouping = df_top5$State
)





# ==========================================
# 5. BUILD THE BOTTOM 5 VISUALIZATION
# ==========================================
df_bottom5 <- df_likert_prep %>% filter(State %in% bottom_5_states)

likert_bottom5 <- likert(
  items = df_bottom5[, "bins_gun_deaths_per_100000_residents", drop = FALSE],
  grouping = df_bottom5$State
)
```

```r
# Apply the custom colors directly inside the plot() function
plot_top5 <- plot(likert_top5, colors = custom_colors) +
  labs(
    title = "Gun Death Rate Distributions for 5 States with Highest Mortality",
    x = NULL,
    y = "Percentage of Years in Each Rate Bin"
```

```r
  ) +
  theme_minimal() +
  theme(
    legend.position = "bottom",
    legend.title = element_blank(), # <--- THIS REMOVES THE WORD "Response"
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.text.y = element_text(size = 11, face = "bold")
  )

print(plot_top5)
```




```r
# Apply the exact same color scale so the two charts match perfectly
plot_bottom5 <- plot(likert_bottom5, colors = custom_colors) +
  labs(
    title = "Gun Death Rate Distributions for 5 States with lowest Mortality",
    x = NULL,
    y = "Percentage of Years in Each Rate Bin"
  ) +
  theme_minimal() +
  theme(
    legend.position = "bottom",
    legend.title = element_blank(), # <--- THIS REMOVES THE WORD "Response"
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.text.y = element_text(size = 11, face = "bold")
  )

print(plot_bottom5)
```



Resources Gun law and death data came from Everytown (WONDER CDC)https://everytownresearch.org/rankings

https://www.cdc.gov/nchs/state-stats/deaths/firearms.html?CDC_AAref_Val=https://www.cdc.gov/nchs/pressroom/sosmap/firearm_mortality/firearm.html